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Data Driven Decision Support System for Predicting Emissions of Ferry Ship

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Abstract

Worldwide trade is a major industry, relying on sea transport but it has significant negative effects on the environment. To tackle this problem, the European Parliament adopted the Emission Monitoring, Reporting and Verification System for Ships (EMRV) in 2018. This system provides an all-encompassing framework for tracking, reporting, and verifying greenhouse gas emissions from organization, company, or governmental level. It is important for global efforts to fight climate change because it helps watching development, points out areas where improvement might be needed and makes sure that reduction targets are met. A Data Driven Decision Support System (DSS) has been developed to predict ferry ship emissions using Automatic Identification System (AIS) data such as position, speed and heading. Holtrop-Mennen method is implemented in Python language to obtain accurate estimation of vessel's resistance thus enabling fuel consumption estimation. Estimated fuel consumptions are used in emissions estimation which considers factors associated with ships fuel type and engine specification. Findings are communicated through Power BI dashboard which enhances accessibility and interpretation. This interactive tool facilitates visualization of key emission metrics, trends and comparisons thereby giving deeper insights beyond numerical data analysis scope. This study offers real time understanding into fuel usage and release hence enabling operators optimize their routes, speeds, and operational parameters for higher effectiveness levels. This thesis aims at establishing a DSS that estimates emissions from ferries to promote sustainable decision making in maritime transportation.

Declaration

I truthfully declare that I have written this thesis independently, that I have completely and accurately indicated all aids used, and that I have marked everything that has been taken from the work of others, either unchanged or with modifications, and that I have complied with the KIT Statutes for the Safeguarding of Good Scientific Practice as amended from time to time.

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List of Abbreviations

AIS	Automatic Identification System
B	Breadth
BI	Business Intelligence
CH ₄	Methane
CO ₂	Carbon dioxide
COG	Course Over Ground
D	Diameter
DSS	Decision Support System
E	Emissions
ECA	Emission Control Areas
EEA	European Economic Area
EEDI	Energy Efficiency Design Index
EF	Emission Factor
EMRV	Emission Monitoring Reporting and Verification System
ETL	Extract Transform and Load
FC	Fuel Consumption
FEU	Forty-foot Equivalent Unit
GHG	Greenhouse Gas
GPS	Global Positioning System
GT	Gross tonnage
HFO	Heavy Fuel Oil
HSD	High Speed Diesel
IMO	International Maritime Organization
ISO	International Organization for Standardization
L	Length
LNG	Liquefied Natural Gas
MDO	Marine Diesel Oil
MDS	Management Decision System
MIS	Management Information System
MSD	Medium Speed Diesel

NO _x	Nitrogen Oxide
OLAP	Online Analytical Processing
OS	Operating Systems
P	Pitch
PM	Particular Matter
r	Radius
Ro-Ro	Roll on-Roll off
S	Wetted Area of Hull
SEEMP	Ship Energy Efficiency Management Plan
SFC	Specific Fuel Consumption
SOLAS	Safety Of Life at Sea
SO _x	Sulphur Dioxide
SSD	Slow Speed Diesel
STW	Speed Through Water
T	Draught
t	thrust deduction coefficient
TEU	Twenty-foot Equivalent Unit
UI	User Interface
UNCTD	United Nations Conference on Trade and Development
UNFCCC	United Nations Framework Convention on Climate Change
V	Ship Speed
w	wake fraction coefficient
Z	number of propeller blades

List of Symbols

L_{WL}	Length on Water Line
L_{OA}	Overall Length
ρ	Seawater Density
∇	displacement
C_B	Block Coefficient
L_{pp}	Length between the perpendiculars
L_{CB}	Longitudinal Center of Buoyancy
C_M	Midship section coefficient
h_B	Center of bulb area above the keel line
C_{WP}	Waterplane area coefficient
A_{BT}	Transverse bulb area
A_T	Transform area of a ship
CP	Prismatic Coefficient
P_{CKL}	Propeller clearance with keel line
R_{total}	The total resistance of a ship
R_F	Frictional resistance
$1+k_1$	form factor
R_{APP}	Resistance of appendages
R_W	Wave-making and wave-breaking resistance
R_B	Additional pressure resistance of bulbous bow near the water surface
R_A	Model ship correlation resistance
R_{TR}	Additional pressure resistance of immersed transom stern
L_R	parameter reflecting the length of the run
θ	the true heading of the ship
θ_c	the current direction of the ship
R_n	Reynold's number
S_{APP}	The wetted area of the appendages
$1+k_2$	The appendage resistance factor
F_n	Froude number

i_E	the half angle of entrance
F_{ni}	Froude number based on the immersion
P_b	Coefficient measure for the emergence of the bow
P_E	Effective Powe
P_B	Brake Power
η_T	Total propulsive efficiency
η_H	Hull efficiency
η_S	Shafting Efficiency
η_R	Rotative Efficiency
η_O	Open water efficiency
η_B	Behind efficiency
ΔW	velocity correction coefficient

1 Introduction

Marine transportation plays a pivotal role in facilitating 80–90% of global trade, transporting over 10 billion tons of various cargoes annually across the world's oceans¹. Over time, sea travel has been a historical means for the movement of people and goods between continents. While air travel became preferred for perishable and high-value items, the industrial revolution and the adoption of containerization significantly enhanced the efficiency of marine transportation. Containerization, exemplified by the standard 20-foot equivalent unit (TEU), revolutionized cargo handling by allowing for seamless inter-modal connections and standardized shipping practices. The international regulation of containers by organizations like ISO further demonstrates their importance. To support the expansion of international trade, the marine transportation industry has consistently sought out bigger and more effective ships, measured in TEUs, or the forty-foot equivalent unit (FEU). This endeavor emphasizes how container standardization has revolutionized marine transportation at the same time fitting with the globalization trend. Furthermore, larger, and faster ships, as examples of technological improvements, have only strengthened the importance of maritime transportation as a cost-effective and reasonable means of international trade in the modern era. These beneficial advances led to an increase in the exchange of dioxides, which led to an increase in emissions.

1.1 Motivation

Understanding and controlling emissions in the maritime sector is crucial given the significant impact of the industry on the environment. According to the Fourth IMO Greenhouse Gas Study, shipping produces about 1,076 million tons of CO₂ each year and is responsible for about 3-5% of global greenhouse gas (GHG) emissions².

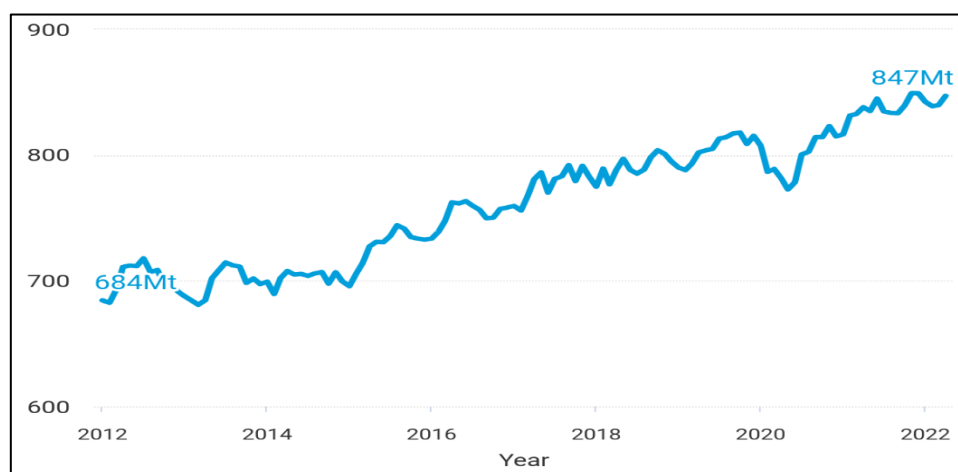


Figure 1. 1 Total CO₂ emissions of the world's merchant fleet, 2012 - 2022 in million tonnes (Source: UNCTD 2022, p. 1)

¹ Walker et al. 2019, p. 1.

² International Maritime Organization 2020a, p. 19.

The numbers had an upward trend from 2020 to 2021, mostly driven by the growth of dry bulk carriers, container ships, and general cargo vessels. The 2022 report from the United Nations Conference on Trade and Development (UNCTD) reveals that the global merchant fleet's total CO₂ emissions increased from 684 million tonnes in 2012 to 847 million tonnes in 2022, as shown in the provided figure. The International Maritime Organization (IMO) estimates a 130% increase in emissions by 2050 in comparison to 2008 levels, emphasizing the importance of addressing the impact on the environment. Another significant environmental concern, in addition to emissions, is the average age of the fleet. Older ships tend to generate higher quantities of pollution. The mean age of ships presently is 21.9 years, determined by the total number of ships, and 11.5 years, determined by their carrying capacity, as depicted in the image below.³.

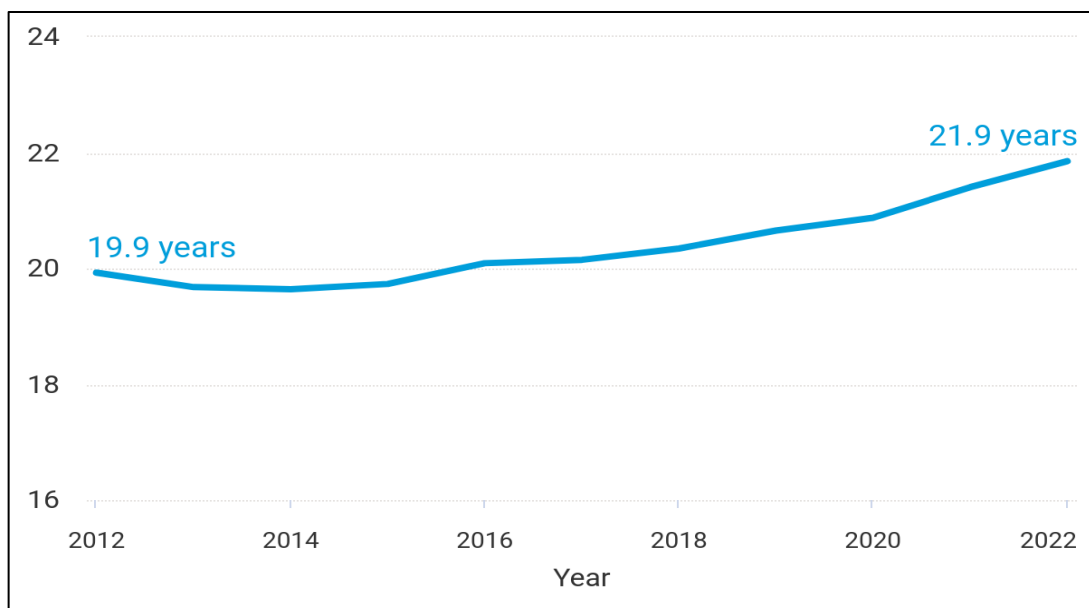


Figure 1. 2 Average age of merchant fleet, 2012-2022 (Source: UNCTD 2022, p. 2)

The objective of the 2015 "Paris Agreement" ratified by the United Nations Framework Convention on Climate Change (UNFCCC) is to restrict the increase in global average temperature to 1.5°C within the next hundred years⁴. Achieving a seamless integration of these difficulties with global initiatives underscores the need of comprehending and mitigating the carbon footprint within the maritime industry.

To significantly reduce greenhouse gas (GHG) emissions from international shipping, it is necessary to implement globally effective legislation. The International Maritime Organization (IMO) implemented the following measures to achieve this objective: Formulate and implement new goals for reducing greenhouse gas emissions and develop and approve a series of measures to achieve these goals:

³ United Nations Conference on Trade and Development 2022.

⁴ United Nations Framework Convention on Climate Change 2015, p. 22.

- To decrease the level of carbon emissions, it is necessary to enhance the energy efficiency of ships by implementing stricter design regulations.
- By 2030, compared to 2008, CO₂ emissions per transport operation in international shipping must be lowered by at least 40%.
- By 2030, it is necessary for international shipping to utilize technology, fuels, and energy sources that emit little to no greenhouse gases. These sustainable alternatives should account for a minimum of 5% and strive for 10% of the total energy consumed in the industry.
- By 2050, net greenhouse gas emissions are expected to be zero.

In addition to these strategies to reduce the amount of CO₂ emitted by the maritime transport sector, in 2018 a regulation came into force which was introduced by the European Parliament establishing the CO₂ Emission Monitoring, Reporting, and Verification (MRV) system⁵.

The primary responsibilities for shipping companies are as follows:

- Companies are required to oversee and track greenhouse gas emissions, fuel consumption, and various other factors. This data is used to generate an annual emissions report, which must be confirmed by an approved MRV shipping verifier.
- Companies must submit a verified emissions report to the appropriate regulatory authority for any ship that was involved in maritime transportation operations inside the Europe during the preceding reporting period, which is the calendar year.

In Practice, a ship operator must monitor and report their respective ship's CO₂ and other emissions like Sulphur Oxides (SO_x), Nitrogen Oxides (NO_x), Particulate Matter (PM), and Methane (CH₄) to the authorities.

1.2 Problem Statement

The challenge in accurate prediction & monitoring of ship emissions that inhibits effective control & mitigation efforts cannot be overstated. For one to give a correct estimate of ship emissions you need to factor in things like ship design, shape, size, speed, and environmental conditions. Several methods can be employed in determining the resistance offered by vessels. This data can be difficult to acquire especially for smaller ships which operate in less regulated areas. Hence this study will focus on using AIS data combined with Holtrop & Mennen's method⁶ to estimate ferry ship's resistance using advanced techniques that will

⁵ European Parliament 2015, pp. 55–76.

⁶ J. Holtrop and G.G.J. Mennen 1982.

enhance accuracy in assessing emissions from ships to have a more inclusive knowledge of environmental impacts caused by maritime transport.

1.3 Research Purpose and Goal

The primary objective of this thesis is to design and implement an advanced Decision Support System (DSS) leveraging cutting-edge technologies, such as Python for data analysis and Business Intelligence (BI) tools for visualization, to predict and optimize emissions like CO₂, NO_x, SO_x, and PM of a ferry ship during its voyage in 2021. Ship operators are given a robust framework for preparing the assessment and having control over discharges by the DSS through the utilization of past records from the Automatic Identification System (AIS) dataset. The Ship's Speed, the Course Over Ground (COG), wind direction and true ship direction are some of the most essential parameters contained in AIS which is used as a basis for forecasting emissions (Methodology). Emissions are estimated using the method of Holtrop & Mennen which is a well-known empirical technique for calculating total ship resistance. Consequently, it becomes possible to predict fuel consumption as well as predicting emissions after determining effective power needed to overcome this resistance. This implies that if ships reduce their speed, then they would be able save fuel since there exists a cubic relationship between ship's speed and power required (providing rationale for the study's emphasis on speed reduction). A Python-based analysis combined with Business Intelligence (BI) visualization tools enables real-time decision-making support for operators of vessels through integration of the method with Holtrop & Mennen approach. A proactive stance adopted here permits ship speed adjustments leading to increased fuel efficiency coupled with lower carbon outputs thus the main purpose of this paper is to promote environmentally friendly maritime operations by providing practical insights using comprehensive data analysis and visualization within our proposed Decision Support Systems (DSS). Not only does it aim at improving fuel efficiency levels, but it also ensures environmental sustainability in maritime sector worldwide part of giving effect to international efforts aimed at reducing adverse effects of marine activities on ecology such as atmospheric pollution caused by burning fossil fuels. In addition, I want my work to be seen as an attempt that addresses the need for informed decisions within shipping companies toward achieving operational effectiveness without compromising environmental safety.

1.4 Thesis Structure

This thesis is split into five principal segments to achieve a logical structure and progression of the information presented as indicated in the diagram below.

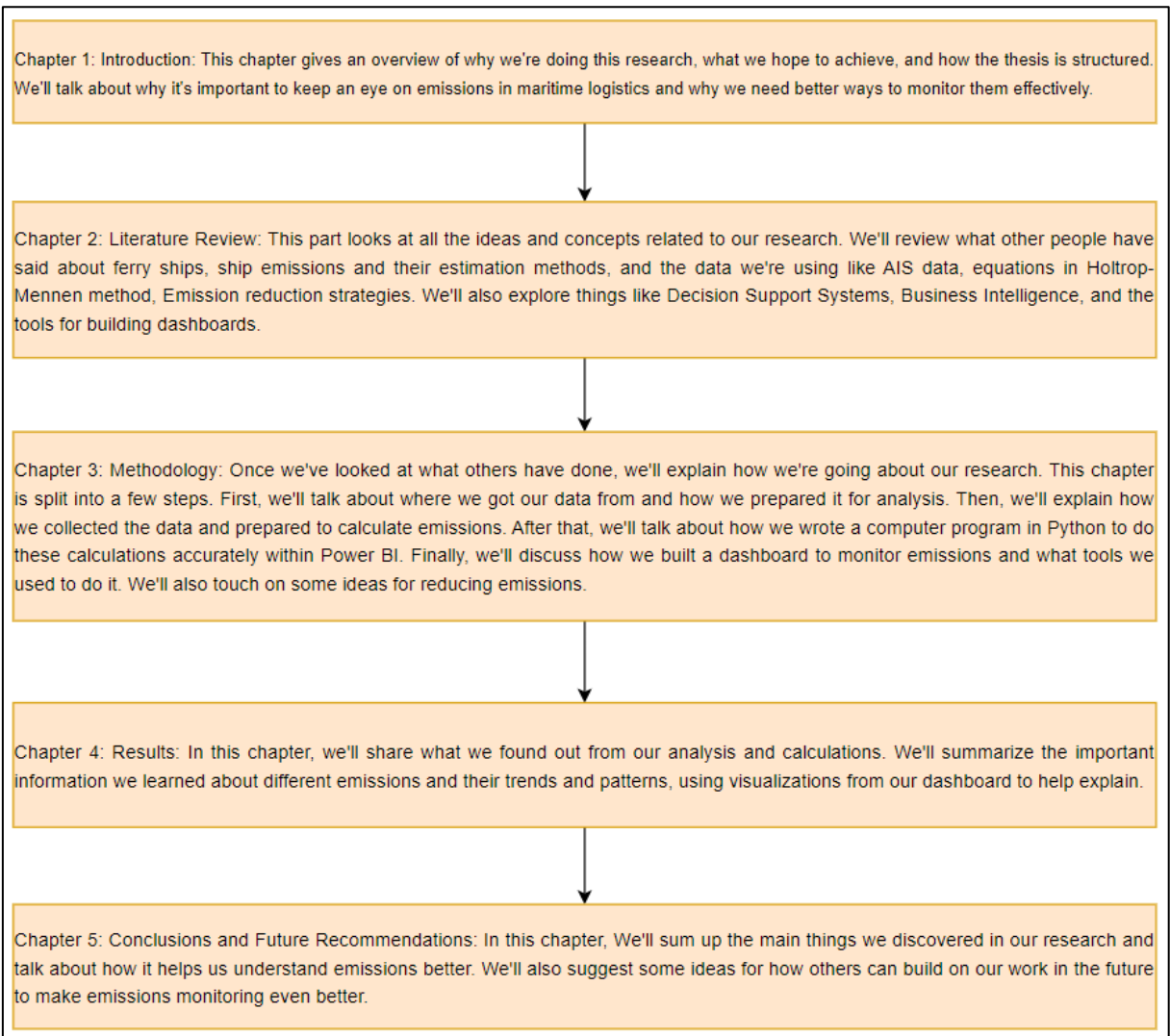


Figure 1. 3 Thesis Structure (Source: Own illustration, 2024)

2 Literature Review

Over the past decade, the substantial economic and environmental impact of maritime operations has led to an increasing interest in improving their sustainability. Over the course of a ship's lifetime, energy systems have the most effects on emissions and energy consumption as well as operational costs. As a result, efforts to enhance the environmental and economic performance of ship energy systems have focused on technology development as well as the exploration of alternative fuels and configurations⁷. One of the technologies that have gained prominence in various fields including ports and the maritime industry is the Decision Support System (DSS) and the focus on developing it has been going on for the last decade.

A study conducted in 2015 focused on developing an artificial neural network-based decision support system to anticipate fuel usage for a single ship utilizing various parameters obtained from Noon Data. The system was designed to be deployed onboard the ship⁸. Similarly, another study was conducted in 2018 to optimize speed by using weather archive data and data mining techniques to understand how weather conditions affect fuel consumption⁹. Despite these research articles demonstrating the importance of DSS in determining and controlling fuel consumption by using various data sources and different tools and methods, research is still ongoing on issues such as estimating a ship's fuel consumption, emissions, and various types of data and methods required for this measurement. The performance and application of a ship are influenced by its size, kind, type of operation, and operating environment. An essential initial step is to simulate ship fuel usage and emissions in actual shipping situations to evaluate the influence of shipping on the climate and the environment, as well as the efficacy of new endeavors targeted at diminishing greenhouse gas (GHG) emissions¹⁰.

A study conducted an analysis of several emission models utilized in the maritime industry and categorized them into two distinct groups: a top-down methodology and a bottom-up methodology¹¹. According to this study, the top-down technique utilizes theoretical models to distribute fuel usage among various ship classifications by merging data on overall fuel consumption acquired via bunker sales. The validity of this approach is currently a subject of ongoing discussion. This approach measures the emissions of individual ships by collecting data on each specific vessel. This method is deemed more accurate as it permits the consolidation of data to approximate the overall emissions from a group of ships. This study

⁷ Yaser. Sharifi et al. 2017, p. 21.

⁸ Beşikçi. Bal et al. 2016, p. 393.

⁹ Habin. Lee et al. 2018, p. 331.

¹⁰ A. Mjelde et al. 2018.

¹¹ Y. Yin et al. 2017.

uses the approach, building upon previous research, to quantify the emissions produced by a ferry ship. Usually, each ship maintains a power reserve to account for weather conditions and guarantee an adequate level of safety. The installed power is not just determined by ship resistance alone. The power three connection is frequently flawed and overly simplistic for the majority of vessels. The power output of the vessel varies precisely based on its type. A 2020 assessment revealed a range of values between 1.5 and 4.8 obtained from various studies and types of vessels¹². As a result, the 'Cubic rule' model is ineffective in accurately predicting engine power output, which hampers the accurate modelling of the effects of new technology and practices that aim to reduce ship resistance or improve ship efficiency. Hence, the model described in this study does not employ the 'Cubic rule'.

To reduce ship collisions, the 2002 International Maritime Organization (IMO) Safety of Life at Sea (SOLAS) Agreement required that all ships weighing over 300 gross tonnages engaged in international trips, and all ships weighing over 500 gross tonnages not involved in international voyages, must have an Automatic Identification System (AIS) installed. The AIS system is responsible for transmitting ship identity, location, and course information. This data allows for a more accurate evaluation of shipping emissions in real operational conditions using an approach that starts at the bottom¹³.

In 2007, a research article was released that pioneered the use of AIS data to estimate emissions. The study employed a load factor to accurately quantify the fuel consumption of each ship¹⁴. However, another study asserts that it is the initial endeavor to forecast emissions in international maritime zones. This study introduces an improved "Cubic Rule" method in their emission model for accurately estimating the fuel consumption of individual ships. A basic speed penalty function is employed to model the effects of waves¹⁵. In 2012, the researchers produced an upgrade to this assessment model to enhance its accuracy. The calm water resistance is calculated using the Hollenbach method¹⁶. The propeller efficiency is determined by use Watson's empirical formula. This method is thought to enhance engine power estimations compared to previous models, especially when the dimensions of the hull are known but the ship's design speed is not known¹⁷. However, the wave model remains unchanged according to another study in which machine learning and physics-based models are combined to predict fuel consumption on big ship fleets¹⁸. The third IMO GHG Study comprehensively examines maritime emissions using AIS data, using both bottom-up and top-

¹² Adland. Roar et al. 2020.

¹³ International Maritime Organization 2017.

¹⁴ James J. Corbett et al. 2007.

¹⁵ J.-P. Jalkanen et al. 2009.

¹⁶ Klaus Uwe. Hollenbach 1998.

¹⁷ David G. M. Watson 2002.

¹⁸ Guo. Bingjie et al. 2022.

down approaches. In 2015, Smestad published a master's thesis¹⁹ that included heuristics for classifying ships based on AIS data. This created several additional opportunities for AIS research. It was made possible to use this heuristic approach for calculating the emissions of a ferry ship in this research. The resistance is calculated by considering the ferry ship's performance characteristics in different operating conditions which is a crucial factor in predicting fuel consumption and emissions.

In addition to ship emissions estimation, in a study importance of emissions monitoring is well explained by developing a sensor web because regulating shipping emissions necessitates collaboration and cooperation among research institutes, non-governmental groups, industry executives, and government authorities²⁰. According to another research paper on ship emission control in Hong Kong, new insights and understanding should be established to allow cross-sectoral involvement and dialog among the major stakeholders in government, industry, and civil society, which could result in the effective implementation of ECA policies²¹. These research papers were reviewed and are taken as references in selecting a dashboard in Power BI to monitor the emissions of the ferry ship.

Many other research publications were reviewed to gain knowledge for creating a Data-Driven Decision Support System for predicting the emissions of a ferry ship.

2.1 Ferry Ship

A Ferry ship is a type of vessel designed to cross a body of water. As a rule, a regular service is established, which means that ferry ships travel between defined ports and always take the same route. Ferries that are seaworthy and travel, for instance, between two nations or even between islands must be distinguished from ferries that only connect a river's two banks. Ferry connections are abundant worldwide, spanning across all continents and occasionally spanning extended or brief lengths.

2.1.1 The History of a Ferry Ship

In the days before engineers could plan, design, and build bridges and tunnels over and under enormous bodies of water, the ferry ship became a necessity. While there are plenty of ferries traveling around Greece and between its islands today, historians generally believe that Greek mythology is the source of the ferry's most well-known historical application. According to mythology, Charon the ferryman conveyed the souls of the deceased across the River Styx²². The first form of ferries consisted of small boats propelled by poles or oars. As ferries evolved,

¹⁹ Bjørnar. Brende Smestad 2015.

²⁰ Zhou. Fan et al. 2022.

²¹ Zhang. Yiqi et al. 2018, p. 50.

²² Kershaw. Daniel 2023, p. 1.

larger flatboats took the place of smaller ones. Some of these boats were outfitted with sails to be used in good weather conditions. Some larger ferries went so far as to employ horses to propel itself. In these cases, the horses would either walk on a treadmill that was connected to paddle wheels or they would circle a capstan that pulled ropes, causing the ferry to move²³.

In 1790, the first steam-powered ferry in the United States was introduced on the Delaware River as the nineteenth century neared²⁴. Although it did not achieve economic success, it paved the way for a new era of ferries that were bigger, swifter, and more dependable. It became evident that both individuals and their vehicles required the means to traverse rivers, lakes, and bays. Ferries evolved to carry not just people, but also horses, carts, and wagons. Similarly, in modern times, individuals drive their cars, vans, motorcycles, and even trucks onto ferries. It wasn't long until the maritime industry evolved, with steam engines being replaced by diesel in all sorts of boats, including ferries.

Ferries are essential in the everyday lives of individuals in many places worldwide. Even with the construction of bridges and tunnels, many people still depend on ferries to get from point A to point B and back. This is especially true in coastal or archipelagic states like the Philippines, Indonesia, Japan, and the UK, where ferries are a necessary mode of domestic transportation between islands²⁵.

In this thesis, we are considering a ferry ship for which the historical data of its voyage in 2021 is available and the name of the ferry ship is “Hammershus”.

2.2 Hammershus A Ferry Ship

The Ro-pax vessel is optimized to carry passengers and cargo between the Danish islands of Bornholm and Sjaelland.



Figure 2. 1 Hammershus Ship (Source: Jackie. Pritchard 2023)

²³ Jones. Susan et al. 1999, p. 232.

²⁴ Ayres. Robert U. 2021, p. 179.

²⁵ Church. Eve 2022.

2.2.1 General Characteristics

Type	Ro-ro passenger ferry ship
Length	158 m
Beam	24.5 m
Draught	5.85 m
Speed	17.7 knots
Capacity	720 passengers

Table 2. 1 General Characteristics of the Hammershus ferry ship (Source: SHIPPAXINFO 2018)

Accurate data is essential for accurately predicting the emissions of a ferry ship or any other type of vessel. Accurate prediction of emissions involves understanding and modelling various factors, and having comprehensive and reliable data is essential for this purpose. There are several types of data sources available which provide diverse information of the vessels that can be crucial for accurate modelling and analysis. The Automatic Identification System (AIS) data is a vital source of information for comprehending the marine activities of vessels, particularly ferry ships. In the subsequent chapter, we will delve into the details of this data source.

2.3 Automatic Identification System (AIS)

The exchange of navigational data between AIS-equipped terminals is facilitated by the Automatic Identification System (AIS), an automated and autonomous tracking technology that is extensively used in the maritime industry. The system facilitates the electronic exchange of both fixed and changing vessel data among AIS-receiving stations, including those on-board ships, on land, or by satellite.

As soon as the AIS data became publicly available, the original perception of its usage underwent a significant change. The International Maritime Organization (IMO) developed AIS as a standardized system to assist port authorities in efficiently managing marine traffic and preventing vessel collisions. However, it soon became evident that its potential applications may span a larger range of maritime-related economic domains.

2.3.1 Types of AIS Data

There are three types of AIS data which are discussed as below.

- **Satellite AIS:** S-AIS gives improved vessel data in isolated locations where terrestrial AIS is not available.
- **Enhanced Satellite AIS:** Enhanced Satellite AIS offers an unparalleled influence on the number of unique MMSIs obtained and the regularity of position updates in HTZ.

- **Terrestrial AIS:** T-AIS enables communication between ships and shorelines within a 40-nautical-mile radius.

The ability to spot trends and create vessel heat maps on international shipping routes makes historical AIS data equally important. Because we can't forecast the future until we comprehend the past, AIS data is also essential for creating intelligent algorithms and machine learning models that automate navigation and fuel use. Hence in this thesis, we are using the historical AIS data of the Hammershus ferry ship during its voyage in 2021 to predict fuel consumption and emissions.

2.4 Holtrop and Mennen Method

This is an empirical approach employed in naval architecture to approximate the overall resistance of a ship's hull. It was developed by Gert-Jan Holtrop and Gerben Mennen in the 1980s. This method is particularly applicable to displacement ships, which move through the water by displacing a volume of water equal to their weight.

This method offers a means to approximate the overall resistance of a vessel by considering different components of resistance, such as frictional resistance, residuary resistance, and wave-making resistance. The method considers various parameters, including the ship's dimensions, speed, draft, and other relevant factors. The fundamental concept of this method is to utilise empirical equations, which are generated from a statistical study of model test data, to estimate the various elements of resistance²⁶. This method is commonly employed during the first phases of ship design to rapidly evaluate the probable resistance properties of a given hull shape.

In this study, this method is used to determine the resistance of the Hammershus ship from given AIS data.

2.4.1 Ship Characteristics for Estimation of Resistance

One can describe the size and carrying capacity of a ship by using either linear measurements or tonnage. Linear measurements involve three essential aspects: length, breadth, and depth. For ships, the fundamental linear dimensions include Length on Waterline (LWL), Breadth (also referred to as Beam or B), and Draught (T). These specific measurements are integral to a ship's hydrodynamic performance, with water resistance being a dominant factor in the overall resistance encountered.

Length (L): The term "overall length" (LOA) refers to the complete measurement of a ship

²⁶ J. Holtrop and G.G.J. Mennen 1982, p. 166.

from one end to the other.

Breadth (B): The breadth or beam (B) refers to whole width of the vessel's hull. In contrast, the Breadth Molded (B_m) is measurement of width taken inside the inner shells of plating.

Draught (T): The Draught molded on Fore Perpendicular (T_F) is the depth of the ship at the forward end, where the perpendicular is located. Conversely, the Draught molded on Aft Perpendicular (T_A) is the depth of the ship at the aft end, where the perpendicular is situated.

Seawater Density (ρ): The density of seawater is subject to variations influenced by factors such as temperature, salinity, and pressure. Typically, seawater exhibits an average density of approximately 1.025 kilograms per liter (kg/L) at the surface, but this value is subject to alteration based on geographical location and prevailing conditions. The density of seawater experiences an increase with decreasing temperature and escalating salinity. In polar regions characterized by colder temperatures and higher salinity, seawater tends to possess a greater density. Conversely, in regions with warmer temperatures and lower salinity, seawater tends to be less dense. It is crucial to recognize that pressure is an additional factor influencing the density of seawater. In the deeper regions of the ocean, where pressure levels are elevated, the density of seawater generally surpasses that observed at the surface.

The dimensions referred to are shown in Fig. 2.2.

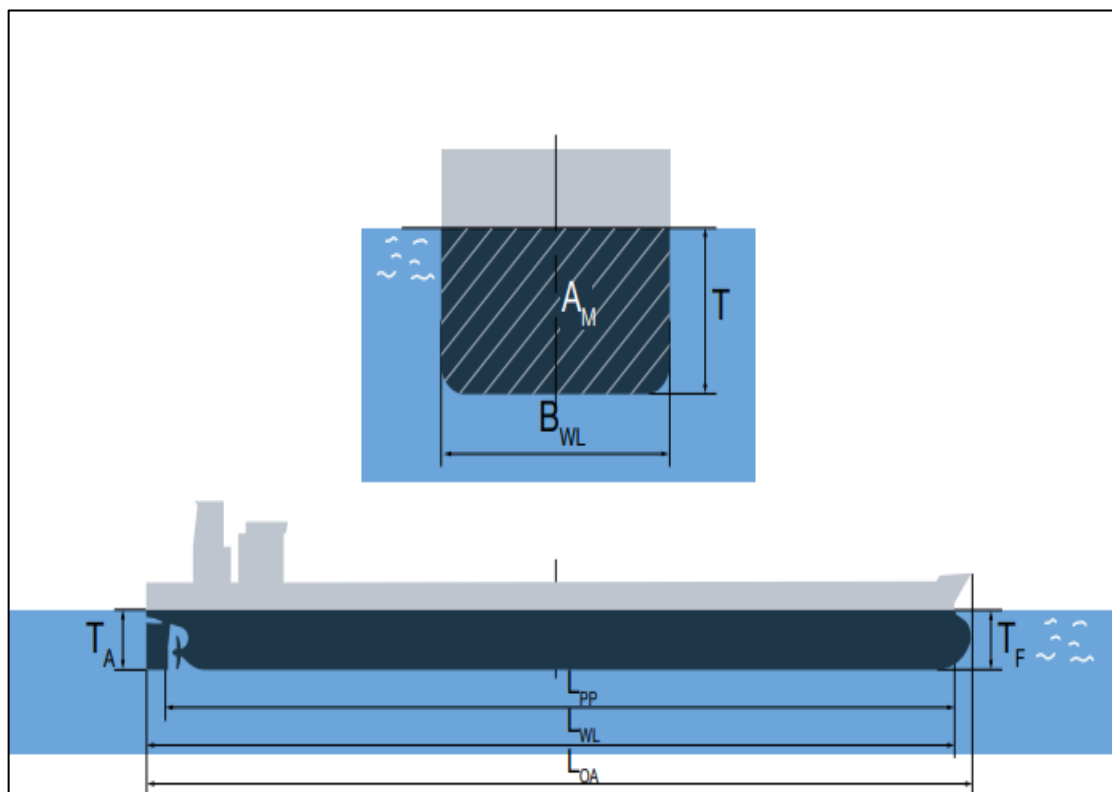


Figure 2. 2 Hull Dimensions(Source: MAN Energy Solutions 2018, p. 5)

Length on Water Line	L_{WL}	144.80 m
Overall Length	L_{OA}	158 m
Breadth moulded	B or B_m	24.50 m
Draught	T	5.70 m
Seawater Density	ρ	1025 kg/m ³

Table 2. 2 Hull Dimensions of Hammershus Ship(Source: SHIPPAXINFO 2018)

Displacement (∇): The term "displacement" in naval architecture denotes the mass of water displaced by a floating ship, measured in units like tons or kilograms. It encompasses not only the weight of the ship's hull but also factors in the mass of cargo, fuel, water, and personnel aboard. Essentially, it quantifies the amount of water pushed aside by the submerged portion of the vessel.

$$\nabla = C_B \times L_{pp} \times B \times T \times \rho \text{ m}^3$$

1

Where:

- C_B = Block Co-efficient
- L_{pp} is Length between the perpendiculars.
- B is breadth
- T is draught
- ρ is density of Seawater

Length between perpendiculars can be determined by following equation.

$$L_{pp} = L_{WL} \times 0.97 \text{ m}$$

2

2.4.2 Hull Coefficients

Longitudinal Center of Buoyancy (L_{CB}): It is a key parameter in naval architecture that describes the longitudinal position of the center of the underwater volume of a ship's hull. It is an important consideration in ship design and stability analysis.

$$L_{CB} = -0.75 \times \left(\left(\frac{L_{WL}}{2} \right) \div 100 \right)$$

3

Midship Section Co-efficient (C_M):

$$C_M = 0.977 + 0.085 \times (C_B - 0.60)$$

4

Centre of Bulb Area above Kneel Line (h_B): For this method, it is stated as a ratio between the ship's draft and itself, as no literature on the subject could be identified. Values from the original Holtrop & Mennen method's case were used to find this ratio²⁷.

$$h_B = 0.4 \times T \text{ m}$$

5

Waterplane Area Co-efficient (C_{WP}): The waterplane area coefficient is determined by the block coefficient at maximum draught and may be calculated using the following equation²⁸.

$$C_{WP} = 0.55 + 0.45 \times C_B$$

6

Transverse Bulb Area (A_{BT}): This can be determined by calculating the ratio between the bulb area and the midship area. Furthermore, it was recommended that 8% of the midship area be utilized²⁹.

It is derived as follows:

$$A_{BT} = 0.08 \times C_M \times B \times T \text{ m}^2$$

7

Transform Area (A_T): The transverse section or transform area of a ship pertains to the configuration and size of the hull's cross-section at a specific location, typically measured perpendicular to the centerline. This element assumes significance in computations related to the hydrodynamic characteristics and stability of the ship. Naval architects engage in a thorough analysis of the transform area to gain insights into the ship's interaction with water, particularly concerning resistance and stability. The variation in hull shapes and transform areas can introduce diverse effects on the ship's overall performance, impacting factors like drag, wave resistance, and stability.

The following equation is used to calculate the transform area³⁰.

²⁷ J. Holtrop and G.G.J. Mennen 1982.

²⁸ Lützen. Marie et al. 2012, p. 7.

²⁹ Rakke. Stian Glomvik 2016, p. 50.

³⁰ Rakke. Stian Glomvik 2016, p. 52.

$$A_T = 0.051 \times C_M \times B \times T \text{ m}^3$$

8

Prismatic Coefficient (C_P): The Prismatic Coefficient (C_P) is a dimensionless parameter used in naval architecture to describe the fullness or fineness of a ship's hull. It is one of the coefficients that provide insights into the shape and hydrodynamic characteristics of the vessel.

The Prismatic Coefficient is calculated using the formula³¹:

$$C_P = \frac{\nabla}{(C_M \times B \times L \times T)}$$

9

The Prismatic Coefficient can take values between 0 and 1. A higher C_P value indicates a fuller hull, while a lower value suggests a finer hull. Here's a brief interpretation of C_P values:

- C_P = 1: This represents a completely box-shaped or prismatic hull.
- C_P < 1: Indicates a fuller or blunter hull.
- C_P > 1: Suggests a finer or slenderer hull.

The Prismatic Coefficient is essential in ship design as it influences various hydrodynamic aspects, including resistance, stability, and maneuverability. Different types of ships, such as cargo vessels, tankers, and fast ships, may have varying C_P values based on their intended purpose and performance requirements. Naval architects use C_P along with other coefficients to optimize the hull shape for efficiency and stability.

Some of the coefficients defined above can be observed as shown in below figure.

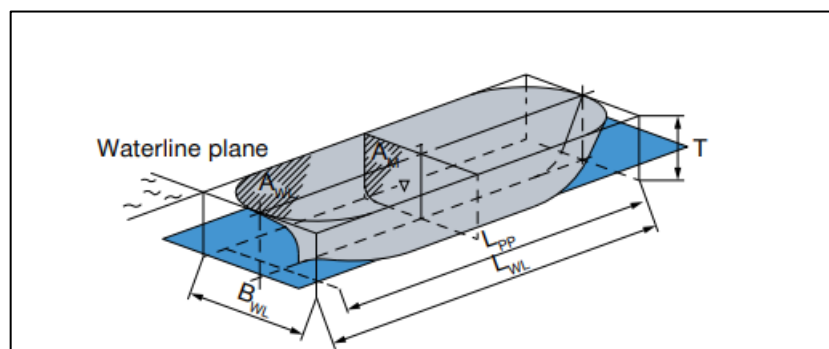


Figure 2. 3 Hull Coefficients (Source: MAN Energy Solutions 2018, p. 6)

Propeller Diameter (D): The propeller diameter, represented by the letter D, is a crucial variable when it comes to ship propulsion systems. It describes the circumference of the circle

³¹ Rakke. Stian Glomvik 2016, p. 49.

that surrounds the outermost tips of the propeller blades when they rotate, and it has a significant impact on the propulsion system's efficiency and overall performance.

$$D = 4 \text{ m}$$

10

The value of D is considered by reference from the Hammershus ship's details³².

Propeller clearance with keel line (P_{CKL}): In ship design and naval architecture, the propeller clearance with the keel line, also referred to as shaft clearance or propeller immersion is a crucial factor. This measurement represents the vertical distance between the keel line, an imaginary line that runs along the bottom of the ship's hull from the bow to the stern, and the lowest point of the propulsion system, which is usually the propeller or propeller hub.

The propeller clearance with the keel line is determined by using the below formula³³:

$$P_{CKL} = (0.48 - 0.02 \times Z) \times \text{Radius} \text{ m}$$

11

Where:

- Z is set to 2 from ship details.
- Radius is half of the propeller diameter which is 2 m.

All the parameters are defined and can be applied to determine the resistance of the ship.

2.5 Resistance Prediction

The force opposing a ship's forward propulsion through the water is referred to as its resistance. A ship's motion is impeded by a variety of obstacles when it travels through water. Ship design and operation depend heavily on the ability to comprehend and forecast these resistances.

The total resistance

$$(R_{total} = R_F (1 + k_1) + R_{APP} + R_W + R_B + R_{TR} + R_A)^{34} \text{ kN}$$

12

where:

³² SHIPPAXINFO 2018, p. 13.

³³ Rakke. Stian Glomvik 2016, p. 54.

³⁴ J. Holtrop and G.G.J. Mennen 1982.

- R_F is frictional resistance.
- R_{APP} stands for Resistance of Appendages³⁵.
- R_W is Resistance against the creation and disruption of waves³⁶.
- R_B is Improved ability of the bulbous bow to withstand high pressure when close to the surface of the water³⁷.
- R_A is Correlation resistance in model ships³⁸.
- R_{TR} is enhanced pressure resistance of a submerged transom stern³⁹.

Form Factor ($1+k_1$):

for the form factor, the prediction formula is taken from Holtrop and Mennen method as shown below:

$$1 + K_1 = C_{13} \left\{ 0.93 + C_{12} \left(\left(\frac{B}{L_R} \right)^{0.92497} \right) (0.95 - C_P)^{-0.521448} ((1 - C_P + 0.0225 L_{CB})^{0.6906}) \right\}$$

13

where:

- C_P is Prismatic coefficient.
- B is Beam
- L_R is the length of the run⁴⁰.
- L_{CB} is the longitudinal point of the center of buoyancy⁴¹.
- C_{12} , C_{13} are coefficients which can be determined according to Holtrop and Mennen method.

The length of the run L_R can be determined by using the below formula⁴²:

$$L_R = L_{WL} (1 - C_P + (0.06 C_P L_{CB}) \div (4 C_P - 1.0)) \text{ m}$$

³⁵ R. van den Berg 2022, p. 20.

³⁶ R. van den Berg 2022, p. 20.

³⁷ R. van den Berg 2022, p. 20.

³⁸ R. van den Berg 2022, p. 20.

³⁹ R. van den Berg 2022, p. 20.

⁴⁰ J. Holtrop and G.G.J. Mennen 1982, p. 166.

⁴¹ J. Holtrop and G.G.J. Mennen 1982, p. 166.

⁴² J. Holtrop and G.G.J. Mennen 1982, p. 166.

The coefficient C_{12} can be determined by using the below formula:

$$(C_{12} = (\frac{T}{L})^{0.2228446} \text{ when } T/L > 0.05$$

$$C_{12} = 48.20(\frac{T}{L} - 0.02)^{2.078} + 0.479948 \text{ when } 0.02 < T/L < 0.05$$

$$C_{12} = 0.479948 \text{ when } T/L < 0.02)^{43}$$

$$(C_{13} = 1 + 0.003 C_{STERN})^{44}$$

The guidelines for C_{STERN} are derived from the table below:

Afterbody form	Cstern
V-shape	-10
Normal shape	0
U -shape	+10

Table 2. 3 Table for Coefficient Cstern (Source: J. Holtrop and G.G.J. Mennen 1982)

Since we are considering the Normal section for our ferry ship,

$$C_{STERN} = 0$$

Frictional Resistance (R_F): Frictional resistance refers to the resistance that arises due to the friction between the water and the hull surface when a ship is in motion.

This can be determined by the following formula⁴⁵:

$$R_F = \frac{1}{2} \rho V^2 S C_F \text{ kN}$$

where:

- ρ is seawater density.

⁴³ J. Holtrop and G.G.J. Mennen 1982, p. 166.

⁴⁴ J. Holtrop and G.G.J. Mennen 1982, p. 166.

⁴⁵ Rakke. Stian Glomvik 2016, p. 16.

- V is the ship's speed through water.
- S is wetted area of the hull.
- C_F is frictional coefficient.

Speed Through Water (STW): It is defined as the actual speed at which a vessel is moving through water. It indicates how fast the ship is progressing in its direction considering the effects of currents and tides. It is measured in knots.

The speed through water can be determined as follows:

$$STW = SOG + SOC_x$$

18

$$SOC_x = SOC \times \cos(\theta - \theta_c)$$

19

where:

- SOG is the speed over the ground.
- SOC is the current speed available from AIS data.
- θ is the true heading of the ship.
- θ_c is the current direction of the ship.

Since the speed through the water is measured in knots, we convert this into meters per second by using the following conversion.

$$1 \text{ knot} = 0.514444 \text{ m/sec}$$

$$S = L (2 T + B) \sqrt{C_M} \left(0.453 + 0.4425 - 0.2862 C_M - 0.003467 \left(\frac{B}{T} \right) + 0.3696 C_{WP} \right) + 2.38 \left(\frac{A_{BT}}{C_B} \right) \text{ m}^2$$

20

The frictional coefficient C_F is determined as follows⁴⁶:

$$C_F = \frac{0.075}{(\log R_n - 2)^2}$$

⁴⁶ Rakke. Stian Glomvik 2016, 20.

Reynold's Number (R_n): The Reynolds number is a basic metric for examining and forecasting the flow patterns surrounding a ship's hull. It is essential to comprehend how the vessel moves through water in terms of hydrodynamics.

The formula for the Reynolds number is:

$$R_n = \frac{V \times L_{WL}}{\text{kinetic viscosity}}$$

Appendage Resistance (R_{APP}): Some ships have appendages such as propellers, rudders, and bilge keels that can contribute to resistance. The design and placement of these components can impact the overall resistance.

$$R_{APP} = 0.5 \rho V^2 S_{APP} (1 + k_2) C_F \text{ kN}$$

(1+k₂) value is determined according to Holtrop and Mennen method:

$$(1 + k_2)_{eq} = \frac{\sum (1 + k_2) S_{APP}}{\sum S_{APP}}$$

$S_{APP} (1 + k_2)$ is the sum of the surface areas of all appendage kinds is multiplied by the resistance factor for each appendage type. The appendage resistance factor and surface area are contingent upon the specific type of appendage. The Hammershus ship is a double-screw vessel. In the case of double-screw vessels, the crucial elements are the balance rudders and the shaft brackets. The shafts, which have a relatively limited surface area, are disregarded in this context. The stabilizer fins are also seen as important for cruise vessels⁴⁷. The resistance factors are stated in Table 2.4

Approximate 1+k ₂ values	
Rudder behind skeg	1.5 – 2.0
Rudder behind stern	1.3 – 1.5
Twin-screw balance rudders	2.8
Shaft brackets	3.0
skeg	1.5 – 2.0
Strut bossing's	3.0
Hull bossing's	2.0

⁴⁷ R. van den Berg 2022, p. 74.

shafts	2.0 – 4.0
Stabilizer fins	2.8
dome	2.7
Bilge keels	1.4

Table 2. 4 Approximate $(1+k_2)$ values (Source: J. Holtrop and G.G.J. Mennen 1982)

For double screw vessels like the Hammershus ship, the $S_{APP} (1 + k_2)$ can be determined as follows⁴⁸:

$$S_{APP} (1 + k_2)_{eq} = (2.8 \times 2 \times 0.015 \times L_{WL} \times T) + \left(3.0 \times 2 \times \frac{1}{2} 0.30 \times T \right) + (50 \times 2.8)$$

25

where:

- L_{WL} is the length on water line.
- T is the draught.

Wave making resistance (R_w): As a ship navigates across the water, it generates waves at both its front (bow) and back (stern). The energy necessary to produce these waves adds to the resistance encountered in wave generation. The form and configuration of the ship's hull have a substantial impact on this resistance element.

$$R_W = C_1 C_2 C_5 \nabla \rho g \exp\{m_1 F_n^d + m_2 \cos(\lambda F_n^{-2})\}$$

26

Where:

$$(C_1 = 2223105 C_7^{3.78613} \left(\frac{T}{B}\right)^{1.07961} (90 - i_E)^{-1.37565})^{49}$$

27

$$(C_7 = 0.229577 (B/L)^{0.33333} \text{ when } B/L_{WL} < 0.11$$

$$C_7 = B/L \text{ when } 0.11 < B/L_{WL} < 0.25$$

$$C_7 = 0.5 - 0.0625 L/B \text{ when } B/L_{WL} > 0.25)^{50}$$

28

C_2 is a parameter which takes into consideration how a bulbous bow's action reduces wave

⁴⁸ R. van den Berg 2022, p. 75.

⁴⁹ J. Holtrop and G.G.J. Mennen 1982, p. 167.

⁵⁰ J. Holtrop and G.G.J. Mennen 1982, p. 167.

resistance and is expressed as follows.

$$C_2 = \exp (-1.89 \sqrt{C_3})$$

29

The coefficient C_5

$$C_5 = 1 - 0.8 A_T / (B T C_M)$$

30

$$F_n = \frac{V}{(\sqrt{g L_{WL}})}$$

31

And the other parameters are determined as below taken from Holtrop-Mennen method:

$$(\lambda = 1.446 C_P - 0.03 L_{WL}/B \text{ when } L_{WL}/B < 12$$

$$\lambda = 1.446 C_P - 0.36 \text{ when } L_{WL} > 12)^{51}$$

32

$$m_1 = 0.0140407 \frac{L_{WL}}{T} - 1.75254 \frac{\nabla^{\frac{1}{3}}}{L_{WL}} - 4.79323 \frac{B}{L_{WL}} - C_{16}$$

33

With:

$$(C_{16} = 1.73014 - 0.7067 C_P \text{ when } C_P > 0.80$$

$$C_{16} = 8.07981 C_P - 13.8673 C_P^2 + 6.984388 C_P^3 \text{ when } C_P < 0.80)^{52}$$

34

And

$$m_2 = C_{15} C_P^2 \exp (-0.1 F_n^{-2})$$

35

With:

$$C_{15} = -1.69385 \text{ when } L^3/\nabla < 512$$

⁵¹ J. Holtrop and G.G.J. Mennen 1982, p. 167.

⁵² J. Holtrop and G.G.J. Mennen 1982, p. 167.

$$C_{15} = 0.0 \text{ when } L^3/\nabla > 1727$$

$$C_{15} = -1.69385 + ((L/\nabla)^{\frac{1}{3}} - 8.0)/2.36 \text{ when } 512 < L^3/\nabla < 1727$$

36

$$d = -0.9$$

37

The half angle of entrance refers to the angle, measured in degrees relative to the center plane, at which the waterline intersects the bow⁵³ i_E .

$$i_E = 1 + 89 \exp \left\{ - \left(\frac{L}{B} \right)^{0.80856} (1 - C_{WP})^{0.30484} (1 - C_P - 0.0225 L_{CB})^{0.6367} \left(\frac{L_R}{B} \right)^{0.34574} \left(\frac{100 \nabla}{L^3} \right)^{0.16302} \right\}$$

38

The coefficient C_3 is:

$$C_3 = 0.56 A_{BT}^{1.5} / \{ B T (0.31 \sqrt{A_{BT}} + T_F - h_B) \}$$

39

Bulbous Bow Resistance near the water surface (R_B): The increased hydrodynamic pressure at the water surface is caused by a complex relationship between the bulbous bow and the flow of water. Generally, the bulbous bow is designed to create a favorable pressure distribution that reduces wave-making resistance and increases the efficiency of the ship's propulsion.

$$R_B = 0.11 \exp(-3 P_B^{-2}) F_{ni}^3 A_{BT}^{1.5} \rho g / (1 + F_{ni}^2) \text{ kN}$$

40

$$P_B = 0.56 \sqrt{A_{BT}} / (T_F - 1.5 h_B)$$

41

$$F_{ni} = V / (\sqrt{g (T_F - h_B - 0.25 \sqrt{A_{BT}})} + 0.15 V^2)$$

42

Resistance of Immersed Transom Stern (R_{TR}): The additional pressure resistance of an immersed transom stern refers to the presence of a transom stern (a flat or nearly flat aft end

⁵³ J. Holtrop and G.G.J. Mennen 1982, p. 167.

of the ship's hull) below the waterline. The transom stern design can have hydrodynamic implications, and understanding the additional pressure resistance is important in ship design and performance analysis. When compared to other stern shapes, such as a traditional cruiser stern or a spoon-shaped stern, a transom stern can result in different flow patterns around the hull. The design of the stern affects the way water flows off the aft end of the ship. The transom stern may lead to the creation of eddies and turbulent flow, which can contribute to increased resistance.

$$R_{TR} = 0.5 \rho V^2 A_T C_6 \text{ kN}$$

43

With:

$$C_6 = 0.2 (1 - 0.2F_{nT}) \text{ when } F_{nT} < 5$$

$$C_6 = 0 \text{ when } F_{nT} \geq 5$$

44

$$F_{nT} = V / (\sqrt{2 g A_T / (B + B C_{WP})})$$

45

Where:

- C_{WP} is the waterplane area coefficient.

Model ship correlation resistance (R_A): It is a process used in naval architecture and ship design to ensure that the hydrodynamic performance predictions obtained from model testing accurately represent the behavior of the full-scale ship. This correlation is particularly crucial when estimating resistance, which includes frictional resistance, wave-making resistance, and other components that oppose a ship's motion through the water. Model-ship correlation is essential because it allows designers to use the results from model testing as a reliable basis for predicting the performance of the full-scale ship. It helps in refining the design and adjusting to achieve accurate predictions of resistance and other hydrodynamic characteristics.

$$R_A = \frac{1}{2} \rho V^2 S C_A \text{ kN}$$

46

With

The correlation allowance coefficient C_A

$$(C_A = 0.006 (L + 100)^{-0.16} - 0.00205 + 0.003 \sqrt{\frac{L}{7.5}} C_B^4 C_2 (0.04 - C_4))^{54}$$

47

With

$$C_4 = \frac{T_F}{L} \text{ when } T_F/L \leq 0.004$$

$$C_4 = 0.04 \text{ when } T_F/L > 0.04$$

48

The coefficient **C₈** is defined as:

$$C_8 = B S / (L D T_A) \text{ when } B/T_A < 5$$

$$C_8 = S \frac{\left(7 \frac{B}{T_A} - 25\right)}{\left(L D \frac{B}{T_A} - 3\right)} \text{ when } B/T_A > 5$$

49

The coefficient **C₉** depends on a coefficient **C₈** defined as⁵⁵:

$$C_9 = C_8 \text{ when } C_8 < 28$$

$$C_9 = 32 - \frac{16}{C_8} - 24 \text{ when } C_8 > 28$$

50

The coefficient **C₁₁** is defined as:

$$C_{11} = \frac{T_A}{D} \text{ when } T_A / D < 2$$

$$C_{11} = 0.0833333 (T_A/D)^3 + 1.33333 \text{ when } T_A / D > 2$$

51

2.6 Power Prediction

Effective Power: In maritime engineering, effective power means the net power that the propulsion system of a ship uses to move the vessel forward by overcoming resistance. It considers many factors that affect the efficiency of power conversion in the ship's engine. These factors include the loss of energy during transmission, resistance of the hull passing through water, and other additional losses due to working conditions and environment.

⁵⁴ J. Holtrop and G.G.J. Mennen 1982, p. 168.

⁵⁵ J. Holtrop and G.G.J. Mennen 1982, p. 168.

Effective power is important as it shows how well a ship's propulsion system works; hence, it directly influences speed, maneuverability as well as fuel economy under different operational conditions.

$$P_E = R_{total} \times V \text{ kW}$$

52

With

- R_{total} = total resistance.
- V = Speed through water.
- P_E = Effective power

Brake Power: Ship's brake power denotes actual output of the propeller or thruster's engine which is calculated with dynamometer. This gauge examines the twisting force and rotational velocity thereby presenting a vivid and definite idea about how well the machine is performing. The power available to carry out external tasks, such moving the ship across the water, is known as brake power. To assess the condition and performance of the ship's engine, Brake Power Monitoring is essential. It is a useful instrument for seeing possible problems, making the most use of fuel, and making sure the engine runs within its designed limits.

The brake power can be determined by:

$$P_B = \frac{R_{total} \times V}{\eta_T \times \text{seamargin}} \text{ kW}$$

53

With

- Sea margin = 0.85
- η_T is total propulsive efficiency.

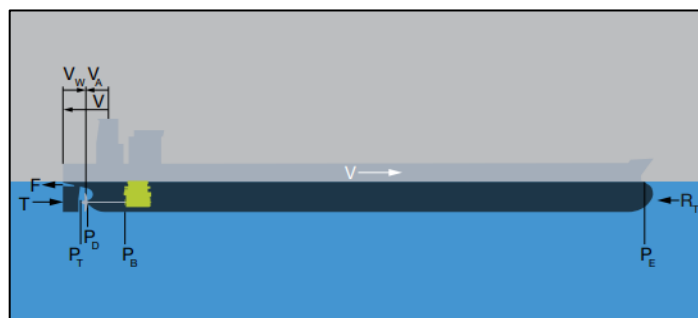


Figure 2. 4 Propulsion Power (Source: MAN Energy Solutions 2018, p. 19)

2.7 Efficiency Components

The total propulsive efficiency is calculated by multiplying the hull efficiency, propeller open water efficiency, relative rotative efficiency, and shaft efficiency⁵⁶.

$$\eta_T = \eta_H \times \eta_R \times \eta_O \times \eta_S$$

54

Hull efficiency can be determined by following formula:

$$\eta_H = \frac{1 - t}{1 - w}$$

55

Where

- t is the coefficient of thrust deduction⁵⁷
- w is the coefficient of wake fraction

The wake fraction⁵⁸ is expressed as:

$$w = 0.11 \frac{0.16}{x} C_B \sqrt{\frac{\nabla^{1/3}}{D}} - \Delta w$$

56

Where

- $x = 2$
- D is the screw's diameter
- Δw = Correction of velocity coefficient⁵⁹

The correction of velocity is contingent upon the Froude number.

$$\text{If } F_n < 0.2 \text{ then } \Delta w = 0$$

$$\text{If } F_n \geq 0.2 \text{ then } \Delta w = 0.1$$

⁵⁶ MAN Energy Solutions 2018, p. 21.

⁵⁷ R. van den Berg 2022, p. 80.

⁵⁸ R. van den Berg 2022, p. 81.

⁵⁹ R. van den Berg 2022, p. 81.

The thrust deduction coefficient⁶⁰ can be expressed as follows for the twin screw vessels:

$$t = 0.8 w (1 + 0.25 w)$$

57

Shafting Efficiency (η_s): Shafting efficiency in ships refers to the effectiveness of the overall propulsion system, specifically the components that transfer power from the ship's engine to the propeller. This critical mechanism guarantees that the engine's rotational power is efficiently transferred into the force required to move the vessel through the water.

Here shafting efficiency taken as 0.99⁶¹.

$$\eta_s = 0.99$$

Rotative Efficiency (η_R): The water flowing behind the propeller has a rotating flow like the hull's wake field, rather than being steady or at right angles to the disk area. The relative rotative efficiency of a propeller affects its efficiency as compared to open water operation.

$$\eta_R = 0.98$$

Open Water Efficiency (η_o): The typical open water efficiency ranges from (0.55 to 0.7)⁶². All vessel types are considered to have an average value of 0.6.

$$\eta_o = 0.98$$

Behind Efficiency (η_B): Behind efficiency refers to the propeller's effectiveness when operating in the wake of the ship.

It is expressed as follows:

$$\eta_B = \eta_H \times \eta_R \times \eta_o$$

58

The next step would be to find the fuel consumption.

2.8 Fuel Consumption

Estimating a ship's fuel consumption requires an understanding of the relationship between the power required for propulsion and the resistance encountered while moving through water. To generate precise estimations, performance monitoring devices that take real-time data on

⁶⁰ R. van den Berg 2022, p. 82.

⁶¹ MAN Energy Solutions 2018, p. 23.

⁶² MAN Energy Solutions 2018, p. 21.

speed, power, and other operating characteristics are widely used. Historical data and performance analysis tools also help shipowners and operators optimize fuel efficiency over time, realizing that correct calculations require thorough information about the vessel.

Since we are considering the Hammershus Ship as vessel with historical data and operating characteristics. The fuel consumption can be determined as follows:

$$FC = SFC \times P_B \text{ g/h}$$

59

With

- SFC = Specific Fuel Consumption

Specific Fuel Consumption (SFC): Specific Fuel Consumption (SFC) is a crucial metric used to measure the efficiency of an engine in converting fuel into useful power. It represents the amount of fuel the engine consumes per unit of power generated. Typically expressed in grams per kilowatt-hour (g/kWh), SFC is a key factor in assessing the fuel efficiency of engines, including those used in maritime and aviation applications. This is a vital consideration for industries where fuel efficiency directly influences operational costs and environmental impact. SFC is determined by dividing the amount of fuel consumed (measured in grams) by the power output (measured in kilowatt-hours) of the engine. The resulting unit is grams per kilowatt-hour (g/kWh), representing the mass of fuel required to produce one kilowatt-hour of power. Understanding and optimizing SFC is crucial for designing and operating engines that strike a balance between performance, cost-effectiveness, and environmental sustainability. It is a fundamental parameter in the evaluation and improvement of engine efficiency across various industries.

Here, SFC for Hammershus Ship is considered as the value shown below which is provided by the ship owner⁶³.

$$SFC = 165 \text{ g/kWh}$$

The next step would be to calculate the emissions of the Hammershus Ship based on its fuel consumption.

2.9 Emissions Prediction

Mitigating ship emissions requires a holistic approach that incorporates regulatory measures, technical developments, and operational improvements. Initiatives such as the International

⁶³ SHIPPAXINFO 2018, p. 5.

Maritime Organization's sulfur cap rule limit the sulfur content in marine fuels, while current efforts focus on improving engine efficiency, researching alternative fuels, and implementing steps in order to mitigate the release of greenhouse gases from the maritime sector. These initiatives aim to establish a balance between maritime operations and environmental sustainability.

The types of emissions are discussed as follows:

Sulfur Oxides (SO_x): SO_x emissions come from burning sulfuric fuels which consist mainly of heavy fuel oil. Consequently, these emissions pollute our air and create acid rains.

Nitrogen Oxides (NO_x): Created when something burns. NO_x discharges cause air contamination and smog to form a significant part of it. As well, they also affect people's health negatively.

Particulate Matter (PM): Comprises tiny particles released from ship engines, including soot and combustion byproducts, impacting air quality and human health.

Carbon Dioxide (CO₂): A greenhouse gas is released through the combustion of fossil fuels, a process that has been found to play a huge role in climate change.

Emissions can be determined as follows:

$$E = FC \times EF \text{ g/h}$$

60

With

- FC = Fuel consumption
- EF = Emission Factor

Emission Factors: Emission factors represent numerical coefficients that describe the amount of a specific pollutant emitted per unit of ship activity or operation. They are indispensable for the prediction and evaluation of the environmental impact of ship emissions. In this respect, they may be regarded and considered as a sign and expression of emissions of specific pollutants, such as carbon dioxide, sulfur oxides, nitrogen oxides, particulate matter, and other pollutants, emitted in the atmosphere per unit of fuel used, distance traveled, or other indicators of the activity measured. Individual types of pollutants have different emission factors, determined by both combustion fuel characteristics and engine technology. The fuel type plays a major role in determining the emission factors, and the fuels are categorized as heavy fuel oil (HFO), marine diesel oil (MDO), liquefied natural gas (LNG), and other alternative fuels. Engine technology and efficiency determine the emission factor because it determines

the combustion process.

In summary, ship emission factors give the quantitative base to evaluate and control the environmental impact of the activities conducted by ships. In other words, they are useful tools for regulatory bodies, policymakers, and the shipping industry to encourage sustainable practices and minimize the ecological footprint of ship emissions.

We calculate the emissions and use the following emission factors for calculating emissions based on fuel and engine type for the Hammershus ferry ship.

Emission Factor CO₂: The CO₂ emission factor is determined by the specific type of fuel utilized, and in the case of our ferry ship, it is Marine Diesel Oil (MDO)⁶⁴, the emission factor for CO₂ is taken as shown in the table below.

The emission factor for CO₂ is:

$$EF\ CO_2 = 3.206\ g / g$$

Fuel Type	EF CO ₂ (g/g)
HFO	3.114
MDO	3.206
LNG	2.750
Methanol	1.375

Table 2. 5 Emission factors for CO₂ for different fuel types (Source: R. van den Berg 2022, p. 85)

Emission Factor SO_x: The SO_x is also based on the type of fuel used like CO₂. The value is taken from the table shown below.

The emission factor for SO_x is:

$$EF\ SO_x = 0.0011027\ g / g$$

Fuel Type	EF SO _x (g/g)
HFO	0.0023774
MDO	0.0011027
LNG	0.0000317
Methanol	0.00264

Table 2. 6 Emission Factors for SO_x for different fuel types (Source: R. van den Berg 2022, p. 87)

Emission Factor NO_x: The NO_x emission factor is determined by the amount of energy used. The age dependency, as represented in the tiers outlined in Table 2.8, and the engine speed

⁶⁴ SHIPPAXINFO 2018, p. 5.

dependency described in Table 2.7, are the factors used to determine NO.

	EF NO_x (g/g)			
Engine Type	Tier 0	Tier 1	Tier 2	Tier 3
SSD	18.1	17	14.4	3.4
MSD	14.00	13.00	10.5	2.6
HSD	10.00	9.8	7.7	2.0
Auxiliary Engine	11.2			
Turbine	4.2			

Table 2. 7 Emission factor for NO_x for all engine types (Source: R. van den Berg 2022, p. 88)

Tier	Construction Year
0	Before 1 st Jan 2000
1	After 1 st of Jan 2000
2	After 1 st of Jan 2011
3	After 1 st of Jan 2016

Table 2. 8 Classification of tiers (Source: R. van den Berg 2022, p. 89)

The emission factor for NO_x is taken by considering the ferry ship in Tier 3 as it is built in 2018 with having Medium-Speed Diesel (MSD) engine type having RPM as 750.

The emission factor for NO_x is:

$$EF NO_x = 2.6 \text{ g / g}$$

Emission Factor PM: The fundamental emission factor for particulate matter (PM) is determined based on energy usage. To ascertain the EF PM, it is necessary to have information on the fuel type, engine speed, and construction year class of the vessel.

Engine Type	Fuel Type	EF PM (g / g)
SSD	HFO	1.39
	MDO	0.19
MSD	HFO	1.39
	MDO	0.18
HSD	HFO	1.39
	MDO	0.18

Table 2. 9 Emission Factors for PM for the construction year class built after 2000 (Source: R. van den Berg 2022, p. 90)

The emission factor for PM is:

$$EF PM = 0.18 \text{ g / g}$$

The emissions Carbon dioxide (CO₂), Nitrogen oxides (NO_x), Sulphur oxides (SO_x), and Particular Matter (PM) of ferry ship has been determined from all the equations derived above.

In the following section, we discuss about the emission reduction strategies.

2.10 Emissions Reduction

This literature study explores different methodologies and technology that have been suggested or put into practice to reduce emissions from ships.

Regulatory Frameworks:

The International Maritime Organization (IMO) has played an important role in setting regulations on ship emissions, which are under MARPOL Annex VI for sulfur oxide (SOx) and nitrogen oxide (NOx) restrictions. The other aspects include the Energy Efficiency Design Index (EEDI) and Ship Energy Efficiency Management Plan (SEEMP), which aim to improve efficiency⁶⁵.

Alternative Fuels:

There is the growing interest in liquefied natural gas as a cleaner marine fuel, which reduces both SOx and PM emissions⁶⁶. Other options under study include biofuels, hydrogen, and ammonia, each with specific challenges and opportunities⁶⁷.

Energy Efficiency Technologies:

Innovative hull optimization, air lubrication systems, and waste heat recovery are some technological improvements in the vessel's energy efficiency to reduce CO2 emissions⁶⁸. Sophisticated engine technology and novel hull coatings significantly improve fuel efficiency⁶⁹.

Retrofitting and Scrubber Systems:

Scrubbers, also known as exhaust gas cleaning systems, are employed to decrease the amount of Sulphur oxide (SOx) emissions produced by ships. This helps ships adhere to rules on Sulphur emissions. However, concerns persist regarding the environmental impact of scrubber wash water discharge⁷⁰.

Slow Steaming:

One strategy to reduce emissions from ships is "slow steaming," which involves operating vessels at lower speeds than their maximum design speed. By reducing speed, ships can achieve significant fuel savings and emissions reductions.

⁶⁵ International Maritime Organization 2021.

⁶⁶ International Maritime Organization 2020b.

⁶⁷ Al-Enazi. Ahad et al. 2021, pp. 1968–1970.

⁶⁸ V J. Jimenez et al. 2022, pp. 7–10.

⁶⁹ International Energy Agency 2019.

⁷⁰ Endres. Sonja et al. 2018.

Emission Control Areas (ECAs):

ECAs enforce stricter regulations on emissions within designated geographical areas, prompting vessels to reduce speed to comply with emission limits. Compliance with ECA regulations necessitates measures such as fuel switching or emission control system installation.

Reducing emissions poses huge challenges for the maritime industry, but strategies and technologies are at hand for mitigating the environmental impacts. Regulatory frameworks, alternative fuels, energy efficiency technologies, retrofitting, shore power systems, and speed management practices contribute to reducing emissions and hence ensuring sustainability in maritime transport. Further research, collaboration, and implementation are key to making meaningful progress toward a greener, more sustainable shipping industry.

The next section is an explanation of the decision support system, types, characteristics, components, and benefits.

2.11 Decision Support System (DSS)

A Decision Support System (DSS) is a computer-based tool that assists in the analysis and decision-making processes within an organization. It facilitates humans in making well-informed decisions by analyzing vast quantities of data and delivering comprehensive information that can be utilized to resolve problems and make decisions. A Decision Support System (DSS) differs from a standard operational application in that it not only gathers data but also aids in making informed decisions. Decision support systems (DSS's) enable enhanced decision-making, prompt problem-solving, and increased efficiency in addressing challenges concerning operations, planning, and management. A Decision Support System (DSS) is utilized by operations management and other planning departments inside an organization to gather information and data, and then integrate it into practical and useful insights. The main objective of utilizing a Decision Support System (DSS) is to offer information to the client in a user-friendly manner. A Decision Support System (DSS) is advantageous as it can be programmed to provide a wide range of reports, all tailored to the individual requirements of the user⁷¹.

2.11.1 History of Decision Support System

The history of Decision Support Systems (DSS) started in the mid-1960s when computerized decision support systems became practicable with the emergence of minicomputers,

⁷¹ García-Alcaraz. Jorge Luis 2023, pp. 170–175.

timeshare operating systems, and distributed computing. The first DSSs were model-driven and built in the late 1960s. Theoretic advances were in the 1970s while financial planning systems, spreadsheet-based DSS, and Group DSS got introduced in the early and mid-1980s. Data warehouses, Executive Information Systems, OLAP, and business intelligence came in the late 1980s and early 1990s. Finally, the chronicle concludes with knowledge-driven DSS and the adoption of Web-based DSS in the mid-1990s⁷².

During the late 1960s and early 1970s, scholars at Harvard and Massachusetts Institute of Technology (MIT) in the United States were the first to introduce the utilization of computers in the process of decision making. Management decision systems (MDS) or management information systems (MIS) referred to computing systems that aided in the process of making decisions. Subsequently, they gained broader recognition under the name of decision support systems (DSS). Scott-Morton coined the term management decision system (MDS) in the early 1970s.

Data warehousing and the internet revolutionized the discipline of Decision Support Systems (DSS) in the mid-1990s. The emergence of the internet led to the development of web-enabled decision support systems (DSS), which have greatly simplified our lives in the present day. Nylund spearheaded the creation of Business Intelligence (BI) software for Procter & Gamble in the late 1990s, with the assistance of Metaphor Computer Systems.

2.11.2 Characteristics of Decision Support System

The major characteristics of a DSS are:

- DSS combines both data and models.
- They are designed to help decision-makers with semi-structured or unstructured activities.
- They improve decision effectiveness rather than decision efficiency.
- They complement management judgment rather than take its place.
- With the use of a DSS, complicated issues that are typically unsolvable with other technological methods can be resolved.
- The main goal of DSS is to give users a very user-friendly interface for interactively manipulating the succession of operations.
- A DSS makes it possible to do a comprehensive, quantitative analysis quickly. It offers

⁷² D. J. Power 2007, p. 1.

quick reaction times to unforeseen events that lead to altered circumstances⁷³.

2.11.3 Benefits of DSS

The benefits of a DSS are:

- enables managers to communicate with each other.
- enhanced performance and control over management.
- DSS-derived decisions are more objective and consistent.
- increases efficacy of management.
- Savings on costs.
- DSS is a tool that can help both individuals and groups⁷⁴.

2.11.4 Components of DSS

The component of a decision support system are as follows:

- Database
- Software Systems
- User Interface

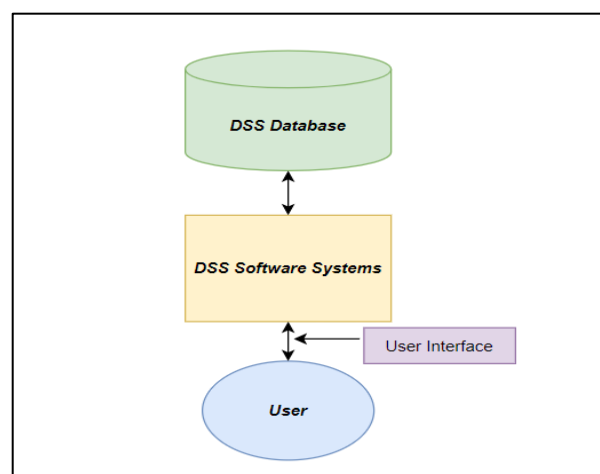


Figure 2. 5 Components of DSS (Source: Own illustration, 2024)

⁷³ Singh. S. K. 2009, p. 583.

⁷⁴ Singh. S. K. 2009, p. 584.

DSS Database: It contains information sourced from every available source, such as external data mined from the internet, data processed by various apps and company-internal data. The database that a decision support system uses can be a small database, a stand-alone one, or a vast data warehouse that provides the necessary information an organization may have. A replica of the production database usually forms part of a decision support system (DSS) database to avoid interference with the operation of operational systems.

DSS Software Systems: It is composed of different analysis and mathematical models that are applied in analyzing complex information and then providing the necessary information. A model identifies the combination of input and conditions to produce the desired output, or it forecasts the output from different input or conditions.

A decision support system may contain many models, and every model has a specific purpose. The user needs, as well as the intended use, defines what a decision support system's family of models includes. In addition, the models that support the types of decisions are predefined by the DSS program.

DSS User Interface: It is the interactive GUI of the DSS that is designed to enable easier communication between the DSS and its users. The findings of the analysis (output) are provided in the forms of text, tables, graphs, or graphics. The user can select the suitable option to see the result, according to his needs.

2.11.5 Types of DSS

In general, there are five different categories of DSS.

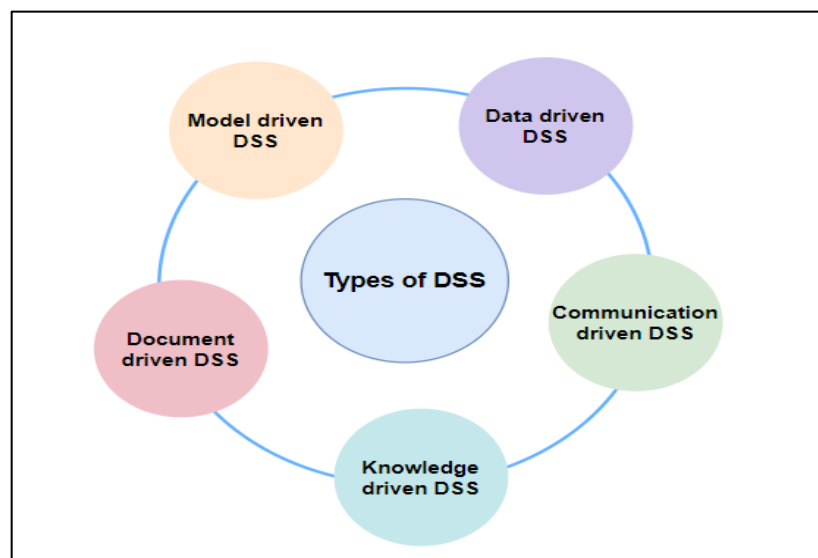


Figure 2. 6 Types of DSS (Source: Own illustration, 2024)

Communication: This decision support system (DSS) allows organizations to facilitate

initiatives that require the collaboration of several personnel. The software incorporates advanced technologies such as Google Docs and Microsoft SharePoint Workspace.

Model: This DSS makes it possible to manage and access organizational, financial, and statistical models. The information provided by users is utilized to collect statistics and establish parameters. The data is integrated into a decision-making model to analyze different possibilities. Dicores is a model-driven Decision Support System (DSS) that is open-source and serves as an example of this kind of system.

Knowledge: This Decision Support System (DSS) utilizes documented data, procedures, regulations, or interactive decision-making models like flowcharts to offer precise and evidence-based resolutions to problems.

Document: This DSS Organizes unstructured data in various digital formats.

Data: A data driven decision support system (DSS) enables users to derive significant insights from extensive datasets to forecast future results. It does this by using computer programs like data mining techniques, which are used to identify bias and trends in data from external or internal databases. Decisions pertaining to financial, commercial, and inventory procedures are primarily made with it. Business intelligence (BI) solutions are the most notable instances of data-driven decision support systems (DSS). Data driven Decision Support Systems (DSS) that utilize Online Analytical Processing (OLAP) exhibit the utmost level of proficiency.

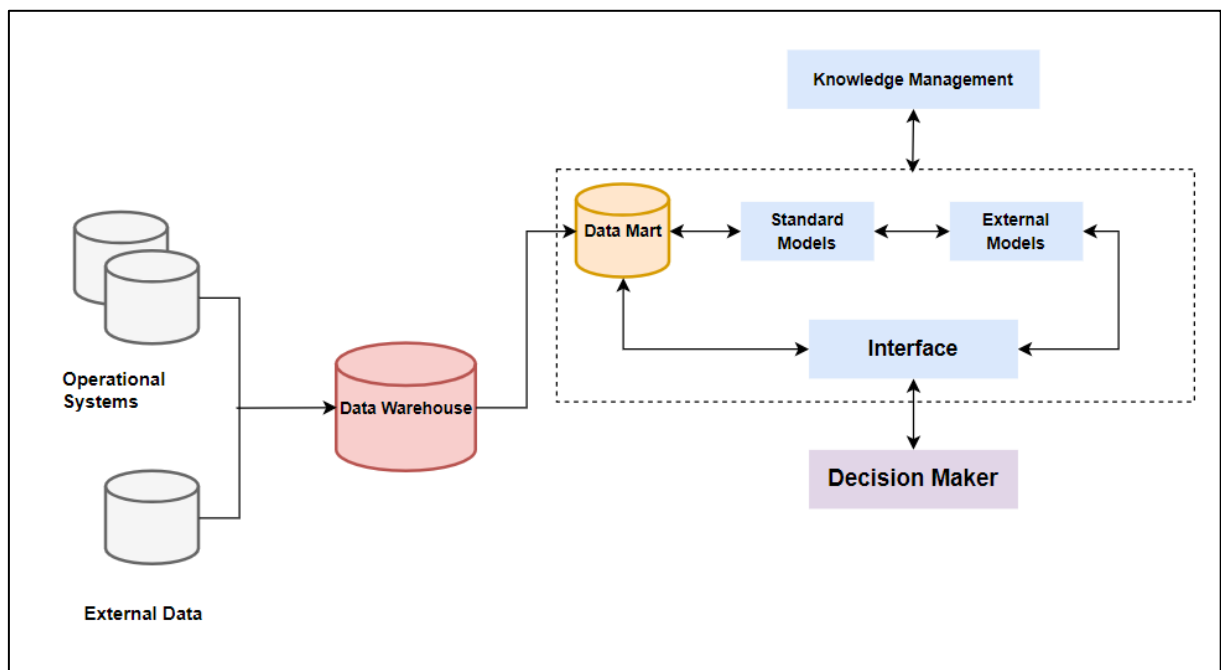


Figure 2. 7 Decision Support System (DSS) Architecture (Source: Own illustration, 2024)

Real-time business intelligence in the logistics industry was enabled by the architecture of business processes to assist decision-making in real-time. Decentralization of decision-making

in the logistics industry has led to the creation of distributed decision support systems, which are increasingly replacing personal decision-making tactics in the corporate environment⁷⁵. In the following section, we discuss about the importance of business intelligence and different BI tools and the selection of a BI tool for visualization emissions of ferry ships aiding decision-making processes.

2.12 Business Intelligence (BI)

Business Intelligence is a combination of technology, tools, and processes that assist companies collect, analyze, and interpret data to make informed business decisions⁷⁶. In any industry, accurate and up-to-date knowledge is essential for making informed business decisions quickly and safely. To address this challenge, business intelligence utilizes big data to provide up-to-date and credible information, enabling companies to monitor their progress and make informed decisions.

2.12.1 History of Business Intelligence

Richard Millar Devens, a writer, used the term "business intelligence" in 1865 when he described a banker who gained market knowledge ahead of his rivals. In 1958, Hans Peter Luhn conducted research on the feasibility of utilizing technology for the purpose of gathering business intelligence. His findings had a significant impact on the formulation of methodologies for constructing IBM's initial analytics systems. To handle and arrange increasing volumes of data, data management and DSS's were created during the 1960s and 1970s.

As per IT education website Data Versity, the current form of business intelligence is believed by many historians to have originated from the DSS database. During that time, several tools were created to retrieve and analyze data and simplify data organization. Several technologies created for DSS implementation including OLAP, executive information systems, and data warehouses⁷⁷.

In the 1990s, business intelligence experienced a surge in recognition, although its technology was intricate and necessitated IT assistance. This frequently led to accumulations and postponed reporting. Even in the absence of information technology, business intelligence analysts and users had to undertake rigorous training to proficiently interrogate and analyze their data. Fortunately, there have been recent advancements that have specifically targeted self-service business intelligence (BI) systems, enabling individuals without expertise in the field to take use of reporting and analysis. Cloud-based solutions have extended the

⁷⁵ Wang. Guozheng 2009, p. 93.

⁷⁶ Howson. Cindi 2013, p. 5.

⁷⁷ Keith D. Foote April 6.

geographical scope of business intelligence (BI). Several current technologies can manage large volumes of data and performing real-time processing, enabling the use of current information for decision-making purposes.

2.12.2 Benefits of Business Intelligence

The benefits of business intelligence in the shipping business are immense, such as improvement in decision-making, increased efficacy, and acting effectively. Some other core benefits are:

- Decision making accurately, informally, and in real-time about historical data.
- Tracking the performance of vessels and fleets for their optimization.
- Fuel consumption, maintenance costs, and operations efficiency analysis for savings.
- Equipment data analysis to predict maintenance needs and reduce downtime.
- Identification and mitigation of the risks associated with maritime operations.
- Optimization of supply chain processes to allow timely deliveries and reduced delays.
- Streamlining of data gathering and reporting for regulatory compliance.
- Better customer service with real-time information and transparency.

2.12.3 Data Flow in Business Intelligence

The very complex dance of business intelligence (BI) commences with the procurement of raw data from disparate sources such as internal databases, spreadsheets, and external repositories. This raw data is then combined using Extract, Transform, and Load (ETL) techniques to unify and coalesce it into a central data warehouse. The next data processing steps ensure the correctness and consistency of data processing, comprising cleansing and transformation. The structured data is then used for data modeling, which identifies linkages and structures.

Analytical tools are used, meanwhile, through statistical and machine learning, patterns and trends are being discovered. Visual representations of these insights can be visualized through data visualization tools, while ad hoc querying and Online Analytical Processing (OLAP) afford dynamic engagement and multi-dimensional analysis.

Data mining technologies uncover hidden patterns and correlations while also contributing to Decision Support Systems (DSS) that offer decision-makers with actionable information. The BI system provides reports and insights, and a critical feedback loop captures user interactions

and preferences to drive continual improvement.

Data governance ensures data integrity, security, and ethical use in addition to user-friendly design features such as UI/UX. Monitoring and maintenance tools keep track of system performance, data accuracy, and user engagement, all of which contribute to the constant evolution of the BI ecosystem. This elaborates an overall data flow in business intelligence, which generates a state of the process with adaptability and iterations that allow firms to source relevant insights and make educated decisions to remain nimble in a fast-changing business landscape.

The data flow in Business Intelligence can be visualized in the following figure:

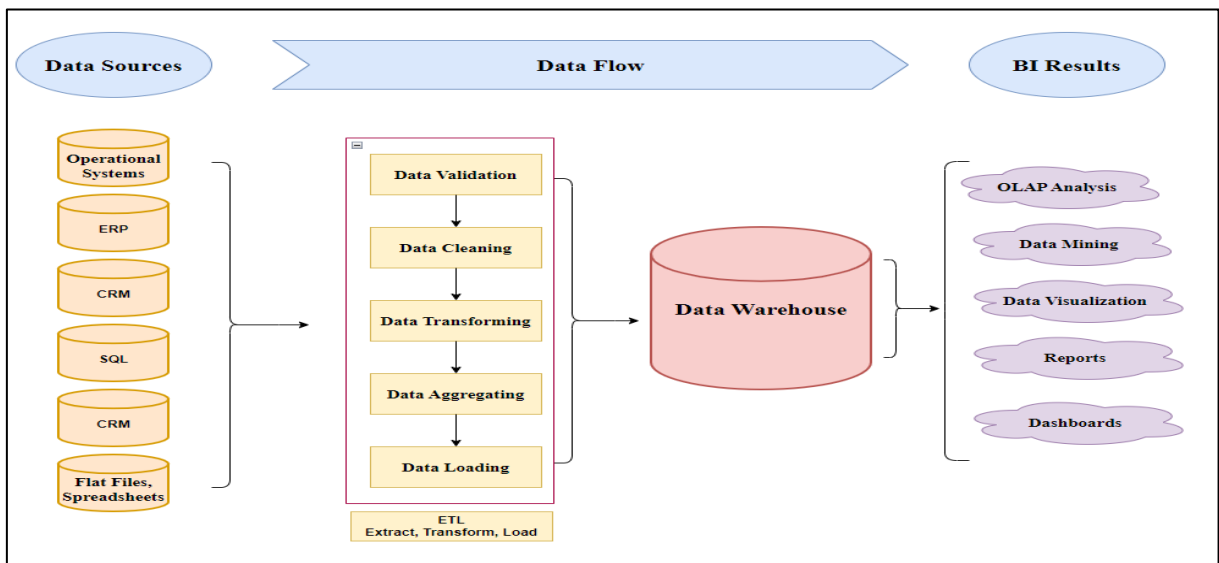


Figure 2. 8 Data flow in business intelligence system (Source: Own illustration, 2024)

2.12.4 Business Intelligence Tools

Business intelligence tools are used to carry out business intelligence, but they do not make decisions on their own. It is important that the results of these tools be integrated with human judgment and intuition, and the information be put into a wider context than can be handled by any data warehouse or knowledge repository. Business intelligence tools are software designed to assist business users in analyzing vast amounts of complex data.

The commonly used BI tools are:

Power BI: Microsoft's Power BI is popular among many industries, including logistics. Its strong features in data integration, modeling, and visualization make it just the tool to analyze logistics data. Power BI can connect to a variety of data sources, including ERP, TMS, and WMS, which enables logistic professionals to keep track of KPIs, monitor performance, and make important insights into supply chain operations.

Tableau: It enables logistics teams to have strong capabilities in data visualization that are very beneficial. This would provide them with the possibility of creating interactive dashboards and reports for real-time visual insight into the most important data: the percentage of delivered orders, the cost of delivery, and the number of units in the warehouse. With intuitive user interaction and high customizability, Tableau is an effective tool for the analysis of logistics data with which one can find the trends and patterns in the given data and identify areas that need further enhancement.

QlikView/Qlik Sense: The BI tool QlikView/Qlik Sense has great, intuitive interfaces with an associative data model. Therefore, these tools are appropriate for logistics people wanting to dive into data relationships, investigate details, and understand supply chain performance. With their interactivity of visualizations and self-service possibilities, the user will be able to analyze logistic data on his or her own and take decisions based on the insights derived from the data.

SAP Business Objects: This is one of the world's leading business intelligence platforms, very well integrated with SAP's logistics and supply chain management solutions. It includes a complete reporting, dashboard, and analytics bundle that enables a logistic professional to analyze data from various sources and draw relevant conclusions. With SAP BusinessObjects, logistics teams can closely monitor key metrics, track progress to goals, and identify areas for further improvement.

MicroStrategy: MicroStrategy is a very advanced business intelligence platform with the provision of very many functionalities to its users for use in reporting, analytics, and mobile BI. This includes performing analysis on logistics data—be it on transportation costs, inventory levels, and order fulfillment. Also included in features are MicroStrategy's data modeling, visualizations, and collaboration that make the solution a must-have for logistics professionals interested in gaining efficiency while operationalizing logistics.

These BI tools come with a variety of features and capacities that can fit the specific requirements of a logistics operation. These tools facilitate organizations in extracting practical insights, enabling them to make well-informed decisions, and enhancing overall efficiency within the supply chain. The choice of BI tool must consider the data sources, integration requirements, user expertise, and specific analytics needs, based on the logistics context.

2.13 Microsoft Power BI Solution

Below are some of the benefits of why Power BI is one of the popular tools for developing emissions visualization dashboards:

- **Compatibility:** Power BI is Microsoft-friendly. Power BI gels very easily with other

Microsoft technologies like Excel, Azure, and SQL Server. If a company has already started working with Microsoft technologies, Microsoft Power BI is just another environment for its employees to analyze and visualize data in.

- **Usability:** It has an easy-to-use interface and drag-and-drop features, making it accessible to all levels of users, from simple to power users. There is a minimum demand for coding in Power BI to create dashboards and reports.
- **Advanced Analytical Skills:** Power BI provides a rich variety of built-in analytical functions and machine learning algorithms using native integration with Azure Machine Learning and other services. This allows for advanced data analysis and forecasting to be done within the dashboard.
- **Data Streaming in Real Time:** Power BI lets you stream data in real time, so you will be able to see and analyze emissions data as it comes through. This is a great way to always be aware of emissions-related events and be ready to respond to them in a timely manner.
- **Performance and Scalability:** With its scalable architecture, efficient query optimization, and broad data connectivity, Power BI can handle large datasets and complex visualizations without compromising performance, even when dealing with extensive emissions data from multiple ferry ships over an extended period.
- **Collaboration and Sharing:** Power BI allows easy collaboration among team members through sharing dashboards and reports. Users can publish dashboards to the Power BI service and share them securely with stakeholders both inside and outside the organization.
- **Cost-Effectiveness:** Power BI has a few pricing options, including a free edition with limited features and subscription-based memberships with more features. Power BI is relatively more affordable compared to other dashboard tools in the market, especially for organizations that are already in the Microsoft ecosystem.

In summary, the features of Power BI are integration capabilities, user-friendliness, advanced analytics, real-time data stream, scalability, collaboration, and affordability; it is a very good solution for making an emissions visualization dashboard for ferry ship operations.

2.13.1 Types of Power BI:

Microsoft offers its Power BI Platform in three distinct services in response to the diverse needs of its customers in creating, using, and distributing business insights, reports, or visualizations:

- Microsoft Power BI Desktop

- Microsoft Power BI Service
- Microsoft Power BI Mobile App

Power BI Desktop: This is a desktop program for users running Microsoft Windows. This is an overall development tool for Power Query, Power Pivot, and Power View. It allows users to create complex queries, models, and reports. Users can easily connect to various data sources, import data, transform it, and visualize the modified data on their desktops. Power BI relies heavily on this product, as it is an essential element frequently utilized in organizations.

Power BI Service: Additionally, it is an internet-based SaaS (Software as a Service) platform that can be conveniently accessed through any web browser. The browser-based version of Power BI offers a comparable environment and user experience to Power BI Desktop, with the added advantage of being available and functional without the need for downloading. Power BI enables users to generate dashboards for their findings and distribute them to others. Additionally, users can build reports directly on the Power BI website. The Power BI Website enables users to analyze and manipulate data online through a web browser, eliminating the need for additional tools.

Power BI Mobile App: The Power BI Mobile Apps are compatible with three mobile operating systems: Android, iOS, and Windows. These applications provide users with an interactive display of reports and dashboards accessed from the Power BI website. Additionally, you have the option to distribute them using your mobile application. One can annotate a portion of the reports and compose a remark regarding it, which can then be shared with others.

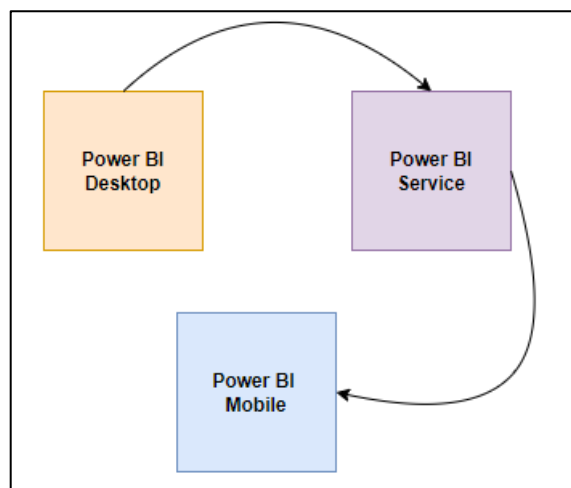


Figure 2. 9 Parts of Microsoft Power BI (Source: Own illustration, 2024)

2.13.2 Elements of Power BI Visualization

The main components that organize or categorize Power BI visualizations are:

Datasets: Power BI's data source refers to the process of importing or connecting to data for gathering. The software provides support for various data sources, a few of which are enumerated below:

Shared Datasets: These datasets in Power BI are standardized, reliable, and optimized. They have been shared by data professionals within an organization to enable other report authors or dataset consumers to connect to them and foster a robust data culture and ethical practices. Any updates or modifications made to the source dataset will have an impact on all reports that are connected to the shared dataset.

Reports: A report in Power BI refers to either a solitary data visualization or a collection of visualizations that represent concealed insights or metrics derived from a foundational dataset. Reports are generated using a consistent data source and can be explored and refined to generate diverse and specific outcomes.

Dashboards: Dashboards consist of a compilation of tiles, reports, or widgets. They are utilized to offer a comprehensive overview of data for business insights and metric analysis.

Tiles: Tiles can be regarded as individual units of data visualizations on your data dashboard. Additionally, it can be generated through the utilization of datasets, dashboards, reports, and several other data sources.

Apps: Applications enable Power BI or traditional BI designers to distribute their content through the App Service. They comprise a set of dashboards and reports that have a common purpose or topic. They are interactive, but end users cannot alter them.

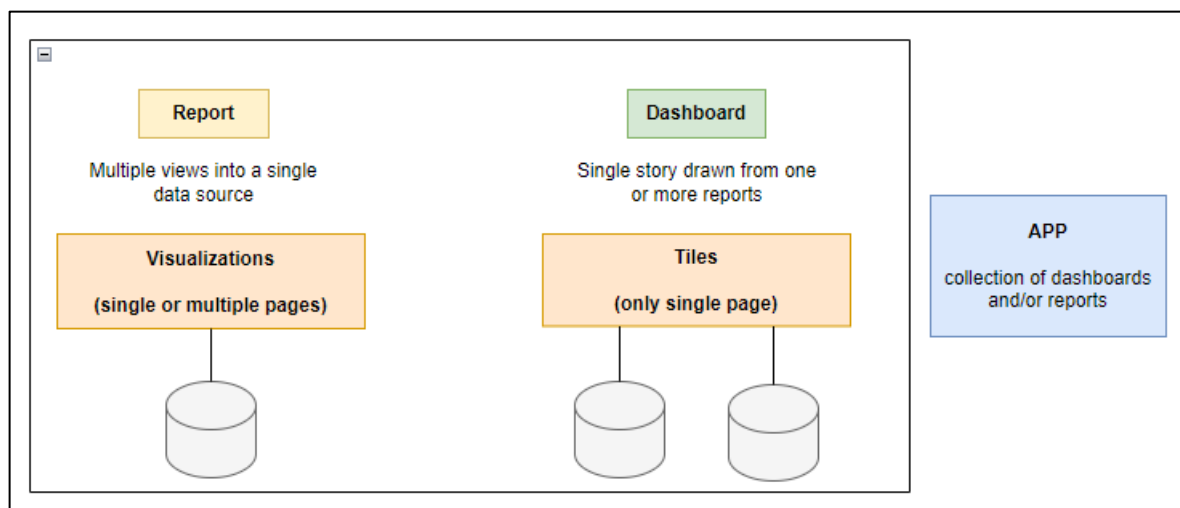


Figure 2. 10 Elements of Power BI Visualization (Source: Own illustration, 2024)

3 Methodology and Approach

3.1 Data Acquisition

Data plays a crucial role in evaluating a ship's emissions. Given our utilization of AIS data to forecast the emissions of the Hammershus ship, it can be sourced from many accessible data repositories. The Danish Maritime Authority website provides historical AIS data in CSV format⁷⁸ for the year 2021.

The data provided includes meteorological information, which is essential for calculating the fuel usage of the ferry ship, as its navigation is affected by weather conditions. In this thesis, we are examining the accurate heading of the ship and the present direction of the ship. The information includes both the current speed and the Speed over Ground (SOG), which are crucial for calculating the speed through water.

The sample Excel sheet which consists of all the parameters in estimating ship's fuel consumption is shown in the following figure.

	P	Q	R	S	T	U	V	W	X	Y
1	SOG	COG	Heading	Width	Length	Draught	Wind Speed	Current Speed	True Wind Directi	True Current Direct
2	14,08729097	293,211204	266,3545151	25	158	5	7,648595184	0,016029888	94,67923401	164,3415184
3	18,14869565	302,2744348	300,1304348	25	158	5	6,700397065	0,104064381	68,88437721	112,8000363
4	10,07736944	163,644294	209,6054159	25	158	5	4,089472207	0,016029888	150,5010888	138,9093824
5	11,43092269	131,5064838	182,0299252	25	158	5	3,821501478	0,016029888	174,2559825	111,3338917
6	18,12529118	126,2925125	125,875208	25	158	5	5,479352572	0,09312816	102,7564163	55,17335839
7	13,03053312	140,4764136	112,9063005	25	158	5	5,38531113	0,016029888	83,55070196	42,210267
8	9,868017058	295,9108742	235,8208955	25	158	5	4,437185899	0,064219864	161,4622886	150,3647574
9	18,33555957	288,2310469	285,7274368	25	158	5	4,675600126	0,016029888	115,0155191	144,9685967
10	18,221673	274,6243346	272,1102662	25	158	5	4,595755766	0,016029888	138,0083795	158,5857673
11	18,21317254	273,6883117	271,6233766	25	158	5	4,378493595	0,080293986	150,3636476	148,9860189
12	17,15072231	298,1372392	296,0016051	25	158	5	2,114902714	0,016029888	133,3922614	134,6944283
13	11,92944882	268,5604724	251,5511811	25	158	5	1,355415102	0,016029888	171,9550894	179,1448524
14	1,420855615	258,9451872	105,0721925	25	158	5	1,369772234	0,016029888	36,69488684	34,37615903
15	14,55101089	118,9449456	118,5645412	25	158	5	1,96541843	0,016029888	32,30861534	47,86850773
16	17,58716745	122,8068858	122,2832551	25	158	4,8	2,035838015	0,016029888	15,4570421	51,5872214
17	18,16945525	101,601751	100,7198444	25	158	4,8	3,24759163	0,016029888	5,09663459	30,02381088
18	18,47833828	98,45845697	97,17804154	25	158	4,8	3,162733033	0,016029888	9,255592231	26,48200806
19	18,37603143	99,06404715	97,92927308	25	158	4,8	3,57508947	0,010751289	2,082416714	7,891289528
20	12,56220339	133,1716949	102,6389831	25	158	4,8	4,083828483	0,016029888	12,21809885	31,94294957
21	0,038582677	340,4102362	174,0708661	25	158	5,04015748	3,334116488	0,054875766	3,347054885	15,36946921
22	15,86049767	288,9043546	285,3328149	25	158	5,3	3,163073579	0,016029888	96,84442799	145,3632188
23	19,01356589	274,0633721	272,6414729	25	158	5,3	3,171711295	0,016029888	68,35499431	158,0545606
24	18,95863192	273,7908795	271,6026059	25	158	5,3	3,975657798	0,068042887	45,19450782	27,81229245
25	18,08745928	282,5868078	280,2084691	25	158	5,3	4,18254414	0,016029888	44,25947916	150,4875644

Figure 3. 1 Input AIS data in MS-Excel sheet (Source: Own illustration, 2024)

The input data for calculating the resistance of a ferry ship using the Holtrop-Mennen method and the fuel consumption from the AIS dataset can be specified as follows:

Speed Over Ground (SOG): The velocity of the vessel with respect to the surface of the earth.

⁷⁸ AIS Data from Danish Maritime Authority.

Current Speed: The speed of ocean currents at the vessel's location

Heading: It is the compass direction the vessel is pointing toward.

True Current Direction: The true direction in which the ocean currents are flowing.

Latitude (LAT): The latitude is the topographical position that indicates the direction from north to south position of a vessel, expressed as degrees.

Longitude (LON): A vessel's longitude is the topographical position that indicates its east-west position, which is expressed in degrees.

Both latitude and longitude coordinates correspond to the route in which the ship is sailing.

3.2 Model Building

Once the data has been gathered in either Excel or CSV format, it is then imported into the Power BI Desktop application. This process is demonstrated in the next part, which is divided into individual steps.

3.2.1 Data

During this stage, the data source is linked to Power BI, namely an Excel sheet containing AIS data for a one-year period.

Data Sources: Power BI Desktop enables you to establish connections with data from many sources.

3.2.2 Python in Power BI

In this project, we are implementing Holtrop-Mennen method in Python Script in Power BI to determine the fuel consumption and emissions. Python is an advanced programming language that is highly recommended for constructing and evolving models. Its popularity stems from its simplicity in coding, support for multiple programming styles, primarily object-oriented and functional programming. Guido van Rossum developed Python and it was first released in 1991. Python programming often entails utilizing an Integrated Development Environment (IDE) to write, test, and debug code. Expanding upon the notion of the Integrated Development Environment (IDE), several tools such as notebooks have been created. Jupyter Notebook is quite popular among users. An open-source Integrated Development Environment (IDE) is utilized for the creation and dissemination of code in diverse programming languages, such as Python, R, Scala, and others. Being a web-based interactive program, it is especially favored by data engineers. Python provides preexisting libraries that facilitate the coding process. These libraries are frequently utilized in the creation of Machine Learning models.

- *NumPy*: NumPy, also referred to as Numerical Python, is a crucial library for doing scientific computations in the Python programming language. It is mostly utilized for performing mathematical computations on data and provides support for operations on objects with many dimensions, known as arrays.
- *Pandas*: This Python module is highly durable and adaptable, mostly utilized for the analysis and modification of data. It seamlessly integrates with NumPy.
- *Scikit-learn*: This Python library is widely utilized in the field of Machine Learning. The development of this software was undertaken by David Cournapeau, a French data scientist, as part of a Google Summer of Code initiative. It includes algorithmic decision-making techniques like as Classification, Regression, and Clustering.
- *Matplotlib*: This is a freely available, multi-platform library that takes inspiration from MATLAB. It enhances the capabilities of NumPy in Python by providing tools for visualizing and graphically representing data.
- *Keras*: Keras is a deep learning application programming interface (API) developed by Google using the Python programming language. It simplifies the execution of neural networks and enables repeated computations of back-propagation neural networks.
- *Tkinter*: This is a Python-specific toolkit designed for creating Graphical User Interfaces (GUIs). It is part of the default Python installation.

Step 1: Importing the Python Script in Power BI to calculate emissions in Excel Sheet

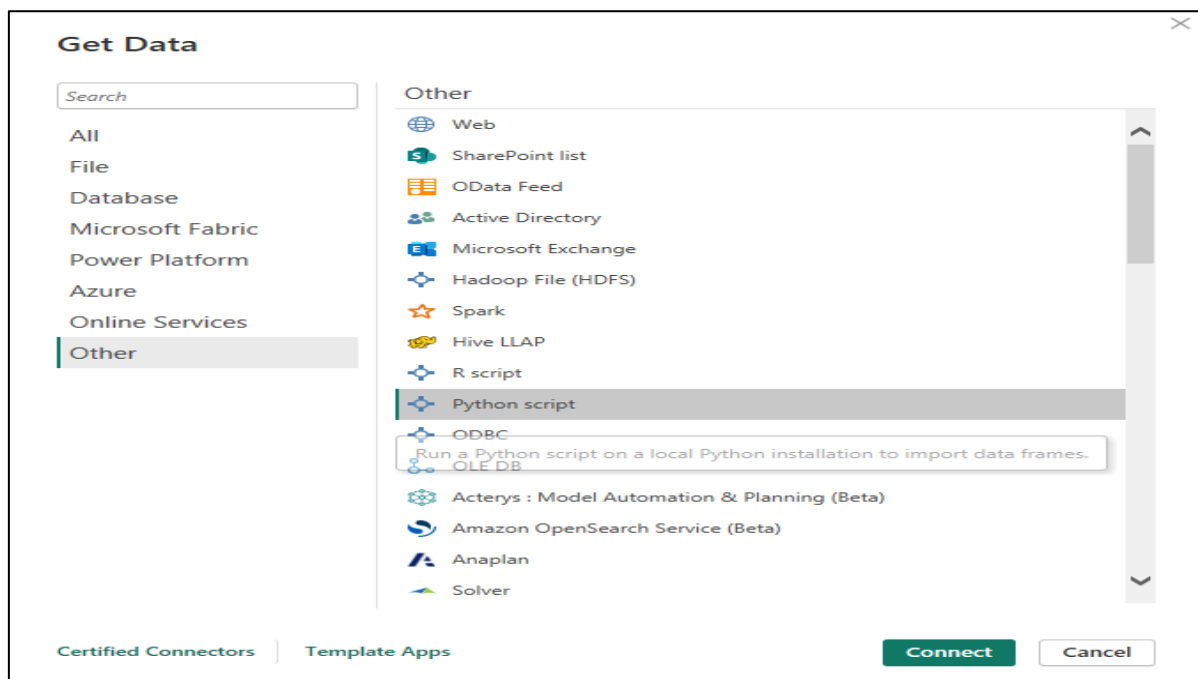


Figure 3. 2 Data Sources in Power BI Desktop (Source: Own illustration, 2024)

Step 2: After establishing a connection between the Python script and Power BI, a navigator panel will appear. In this panel, the Holtrop-Mennen method is implemented as Python code to compute emissions. Once the Python script is executed successfully, the new data will be imported into Power BI. Subsequently, this data is utilized for analysis and visualization, as depicted in the figure below.

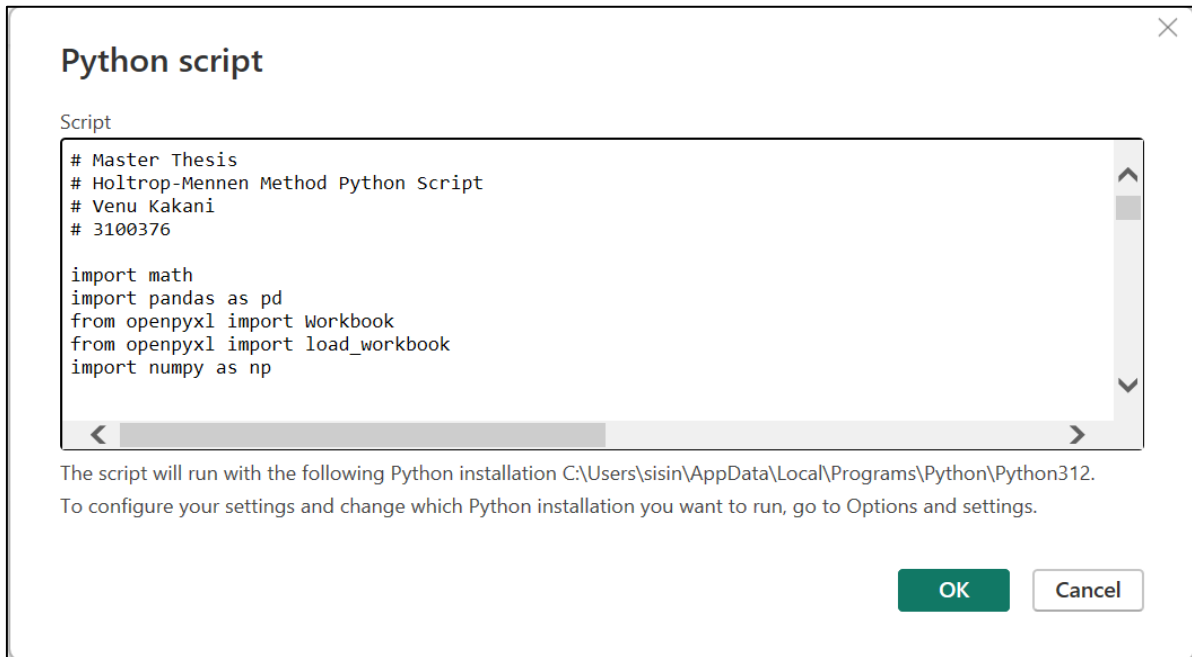


Figure 3. 3 Python Script in Power BI Desktop (Source: Own illustration, 2024)

3.2.3 Data Transformation

The subsequent action is choosing the "transform the data" option to enhance the analysis by cleaning, refining, and arranging the raw data. Converting raw data into actionable insights is a crucial and significant process. The fundamental elements of data transformation in Power BI include:

Power Query Editor: This tool is used for transforming and manipulating data in Microsoft Power BI. It acts as a gateway for retrieving, cleansing, and transforming data from many sources before it is visualized in Power BI reports and dashboards. It provides a wide range of tools, from deleting duplicates and resolving missing information to combining tables and constructing calculated columns.

Step 3: In the power query editor, we filter the empty values or zeros in the column SOG in our dataset to avoid errors in the estimation of resistance and fuel consumption of ferry ship. The power query editor is displayed in the following figure:

serial	sea surface temperature	Combined wind waves and swell height	swell height	Wind wave height	Surface pressure	SOG	COG	Heading
1	3.75762091	279.699583	0.86542307	0.523892238	0.801356177	100138	14.08728097	293.211204
2	3.80910203	279.8203125	0.785674632	0.452493547	0.578048706	100662.73	18.14889363	302.744548
3	3.82600213	null	0.55655568	0.55655562	0.256454957	10021.702625	10.07705944	101.564294
4	3.82054615	null	0.618782303	0.555404902	0.22618203	100184.1875	11.43062369	111.5094638
5	3.815267563	279.8203125	0.67490861	0.598373413	0.263031006	100680.8125	18.12529138	126.2925125
6	3.84317111	279.699583	0.664935332	0.611022949	0.254058838	100555.875	11.00033322	140.4764136
7	3.82377477	279.8583984	0.599344671	0.562347412	0.190124512	100527.0625	9.86801706	295.9108742
8	3.664199829	279.8203125	0.519174814	0.47389355	0.175598145	100662.825	18.19351957	188.2110489
9	3.45087425	279.7102171	0.48460309	0.407571701	0.155813963	100067.6875	18.2156575	274.6142546
10	3.144148827	279.506958	0.476178438	0.358428134	0.297042947	100630	18.21517254	279.6881117
11	null	null	null	null	100653.3125	17.15072232	298.732292	268.5604724
12	null	277.8117871	null	null	100521.25	11.92944882	188.5604724	258.9451872
13	null	278.3225088	null	null	100571.0625	1.400853615	258.9451872	118.9494936
14	3.79981834	279.6987389	0.219887274	0.219191494	0.807506797	100771.3129	14.31010989	118.9494936
15	3.212711242	278.6567389	0.201769586	0.199554442	0.013910015	100781	17.58736740	112.8608618
16	2.87033844	279.5458884	0.260693421	0.251126933	0.11671875	100891.4175	18.16945125	101.601751
17	2.850141404	279.8006592	0.162905141	0.302001953	0.185302754	100896.8175	18.47803826	98.4545697
18	3.298418335	279.8120117	0.419300238	0.380310059	0.143768311	100962.875	18.17803143	98.4545697
19	3.519798279	279.8989238	0.518771333	0.482051664	0.160786802	100921.375	12.16220339	111.1718949
20	3.18269793	280.0805481	0.13885037	0.113491821	0.097915649	101404.25	0.08862877	140.4101382
21	3.02474396	279.8989238	0.294076085	0.281790167	0.079083038	101549.75	11.86049767	188.9045146
22	2.77642809	279.8120117	0.210009374	0.210641488	0.048858843	101694.375	19.01596389	174.0633721
23	2.416842503	279.6984242	0.229313154	0.168548399	0.140487671	101675.625	18.95881392	177.708795
24	2.421658609	279.5458884	0.142383805	0.160891771	0.179931641	101758.125	18.08749328	282.5888078
25	2.228969574	278.6567389	0.17043698	0.141723633	0.086029053	101774.1875	16.69608483	302.1399674
26	null	278.3225088	null	null	101690.5625	2.862007465	2.861705357	181.7501837
27	null	278.3210449	null	null	101760.5625	7.891248379	181.7501837	181.7501837
28	2.16308891	278.5863037	0.159840703	0.130076885	0.081848345	101831.5625	17.33567793	112.8755592
29	2.218284396	279.340332	0.246271233	0.123785089	0.112890625	101960.375	18.51728968	105.5899209
30	2.41650488	279.4056396	0.272533635	0.111191933	0.248776297	101954.9375	18.89498747	96.6769424
31	2.488410141	279.6300049	0.279138894	0.110846107	0.181128652	101995.0625	19.04462759	98.9602882
32	2.695272396	279.7451261	0.454677799	0.115689204	0.414676954	101939.5	12.03060206	111.5177813
33	3.355424881	279.7451261	2.21243161	0.482940674	2.158123195	102121.5625	16.21524059	176.7544262
34	5.424964905	279.6300049	2.132545524	0.564088214	2.054423828	102234.4375	19.2806005	280.0538462

Figure 3. 4 Power Query Editor (Source: Own illustration, 2024)

Step 4: The next step is to apply changes and load the emissions data from the Power Query editor to the Microsoft Power BI Desktop to create Emissions Dashboard.

3.3 Emission Dashboard Development

Building an emissions tracking dashboard is vital for companies that want to measure, interpret, and mitigate their impact on the environment. This chapter explores creating monitors, covering components and key things you should think about when designing one.

The three foundational views when creating a dashboard in Power BI desktop are report, table, and model views. These are pivots in the development of a dashboard that will help users visualize, analyse, and comprehend emission data effectively.

3.3.1 Power BI Views

Report View: The report view in Power BI is where users design and create interactive visualizations based on their data. They can easily build charts, graphs, and tables by dragging and dropping fields onto a canvas. Additionally, users have the ability to enhance their data exploration experience by adding slicers, filters, and other interactive elements.

Table View: On the other hand, the table view presents the underlying data tables and their contents in a tabular format. This view allows users to perform various tasks such as viewing, editing, and managing the data loaded into Power BI. Some specific actions that can be done in this view include renaming columns, adding calculated columns, and applying data transformations.

Model View: Lastly, the model view provides a visual representation of the data model in

Power BI. It displays tables, relationships, and calculations that have been defined within the data model. In this view, users can build and manage relationships between tables, establish hierarchies, and examine table characteristics.

In summary, data modeling in Power BI structures data for analysis, leveraging DAX for calculations, measures, and calculated columns to get insights, and using various views such as Report, Model, and Table to build visualizations, manage data relationships, and view data tables respectively.

3.3.2 Data Modelling

Data modelling in Power BI is the procedure that involves the structuring and arrangement of data in an organized way to make it possible to analyze and report. The process involves importing data from many sources, establishing connections across tables, and creating calculated columns and measures to extract valuable information.

Advantages of Data Modelling: Power BI makes effective use of data modelling, which is very important for many things. This permits users to model different data sources integrated into a single view, implement complex analysis, and properly visualize the data. More importantly, it supports dynamic reporting for insight across different dimensions.

DAX (Data Analysis Expressions): The formula language of Power BI is DAX. It supports creating measures, calculated columns & tables to get value out of data. Behaviour DAX makes it possible to perform complex calculations such as the summation of the values or the filtering the data to obtain the results of patterns over time as smooth as doing reasoning in human language.

Measures: Power BI measures are a way of aggregating data as well as providing actionable insights. It is often used with formulas to perform calculations on numbers like totals of sales, averages, and percentages etc. DAX expressions for writing Measures which can be used for dynamic data visualization.

Calculated Columns: Calculated columns are like extra columns in a table, but they're based on special formulas instead of just holding data. Unlike measures, these calculated columns aren't just for totals or averages across a whole table. Instead, they do their math based on the values in other columns in the same table, row by row.

This lets you make new attributes or do calculations that need to happen at that level.

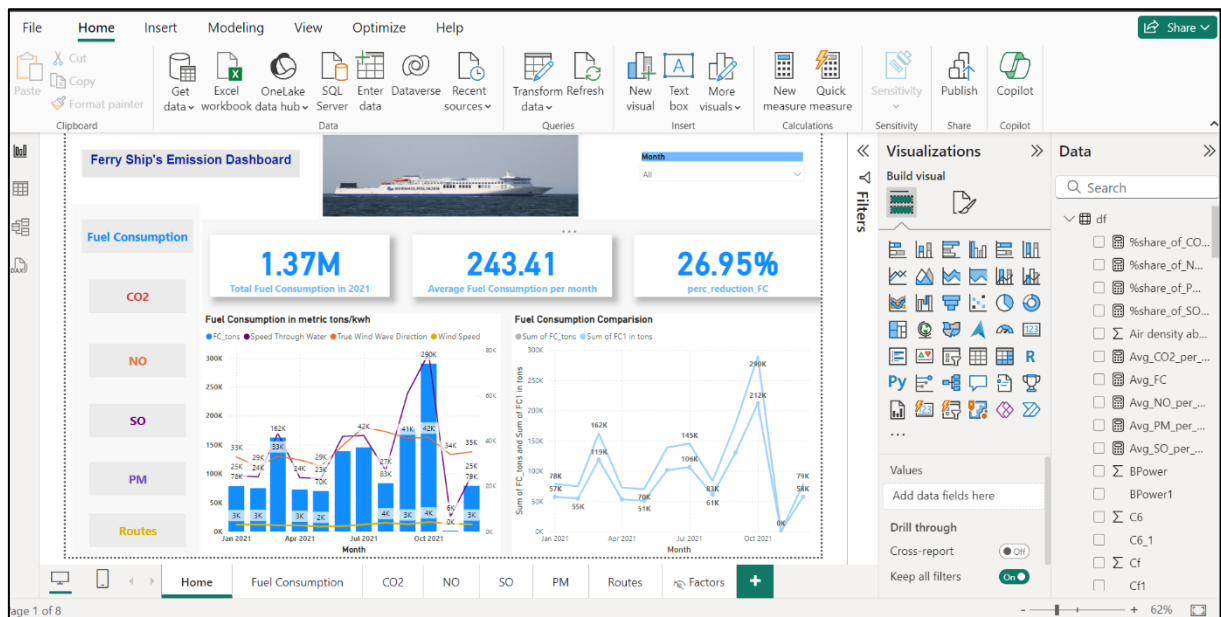


Figure 3. 5 Desktop View in Power BI (Source: Own illustration, 2024)

3.3.3 Emission Calculations

Total Emissions: Total emissions refer to the sum of all the pollutants or greenhouse gases released into the atmosphere for a specified period or in each place. This includes all sources contributing to environmental pollution.

Emissions can be typically measured in mass, that is, kilograms or tons, or in volume, that is, in cubic meters, and may contain various pollutants such as carbon dioxide, nitrogen oxides, Sulphur dioxide, and particulate matter.

Average Emissions: Average emissions indicate the average level of emissions of greenhouse gases released per unit of activity, distance, or time. The information indicates the normal environmental impact of a given process, operation, or form of transport.

Average emissions are determined from the ratio of the total emissions produced over a specific time or activity to the equivalent measure of activity or production. This method results in a standardized measurement that can be applied across the board for comparisons and the evaluation of environmental performance.

Average Fuel Consumption: It is the mean rate of fuel being used per unit of distance travelled, time in operation, or work being performed by a vehicle, machinery, or an industrial process.

The fuel consumption is normally measured in quantities like liters per kilometer (L/km), gallons per hour (gal/hr), or liters per hour (L/hr), among others.

3.3.4 Relationship between Speed, Fuel Consumption, and Emissions

Relation between Speed and Fuel Consumption:

The speed of a vessel is a very significant variable that will affect the quantity of fuel consumed, as generally, higher speeds will have to burn more fuel to sustain. Fuel consumption and speed depend on many variables, including hull drag, engine efficiency, and characteristics of the propulsion plant.

$$FC \propto V^3$$

where:

FC represents fuel consumption in metric tons per hour.

V represents the velocity of the ship as it moves through water, measured in knots.

- The correlation between speed and fuel consumption is commonly represented as cubic, implying that fuel consumption experiences a substantial increase as speed increases.
- Greater velocities lead to heightened hydrodynamic drag, necessitating a greater amount of energy (and fuel) to counteract this drag and sustain speed.
- Factors such as hull shape, propulsive efficacy, and operational concerns can add to the fact that speed and fuel consumption are not directly proportional in their relationship.

Relation between Fuel Consumption and Emissions:

A ship's fuel consumption directly affects its emissions because the combustion of fuel results in emission of pollutants and greenhouse gases.

Emission factors: These are fixed based on the engine's design and type, and the fuel makeup. They specify how many emissions are created per volume of fuel consumed.

$$E \propto FC$$

Where:

E = Emissions of a ship in metric tons per hour

FC = Fuel consumption in metric tons per hour

- Relation between the fuel used and emissions is usually proportional, meaning as the fuel consumed increases the emissions are proportional or linear in nature.
- Emission factors measure the environmental consequences linked to the combustion

of fuel, offering understanding of the emissions produced for each unit of fuel used.

- By reducing fuel consumption through efficient operation and technology adoption, emissions from ships can be minimized.
- Balancing speed requirements with fuel efficiency is essential for minimizing both fuel consumption and emissions.
- Slow steaming, also known as running at reduced speeds, is one of the most applied means of achieving fuel efficiency and lowering emissions, particularly during periods of low cargo demand or in environmentally sensitive areas.

Haversine Formula: The Haversine formula is an important thing in geospatial analysis, namely the calculation of distances between geographical places on the Earth's surface. The formula was originally developed by R. W. Sinnott in 1958 and is grounded on ancient Greek trigonometric concepts. Based on the principles of spherical trigonometry, it allows for the calculation of the shortest distance between two places on the surface of a hypothetical spherical Earth. Of course, its value cannot be overemphasized for history and broad use in navigation and geographical research, and yet its inherent limitation must be considered. The Haversine formula is quite an accurate approximation for small distances but becomes less accurate for greater distances because the Earth is an oblate spheroid, and not a perfect sphere. Therefore, while extremely useful for various practical purposes, users should be careful and explore alternative approaches to exact computations over large distances or with extremely high precision.

d

$= R$

$$\times 2 \tan 2 \left(\sqrt{\sin^2 \left(\frac{\Delta \phi}{2} \right) + \cos(\phi_1) \times \cos(\phi_2) \times \sin^2 \left(\frac{\Delta \lambda}{2} \right)}, \sqrt{1 - \sin^2 \left(\frac{\Delta \phi}{2} \right) - \cos(\phi_1) \times \cos(\phi_2)} \right)$$

This thesis utilizes the haversine formula to calculate distance between two coordinates, and through these results, the number of trips in a month. This data is later applied in measuring emissions and fuel consumption for each specific trip.

3.3.5 Emission Dashboard as User Interface

This Ship Emissions Dashboard in Power BI serves as an end-to-end interface in our Decision Support System (DSS) whereby stakeholders get real-time, fact-based insights on emissions produced by the ferry ship. This interactive dashboard consolidates information on the usage of fuel, speed, and emissions, and allows users to explore trends, identify inefficiencies, and use such data to improve eco-efficiency. With user-friendly visual representations and filters

that can be adapted to users' needs, they may explore the emissions from the ships, for example, according to the routes, time ranges, and obtain important knowledge on carbon footprint, concentrations of pollutants, and compliance with the norms. The dashboard facilitates evidence-based decision-making by providing insight into key performance indicators and actionable measures, thus enabling stakeholders to run programs that can reduce emissions, increase fuel efficiency, and improve sustainability in our maritime operations.

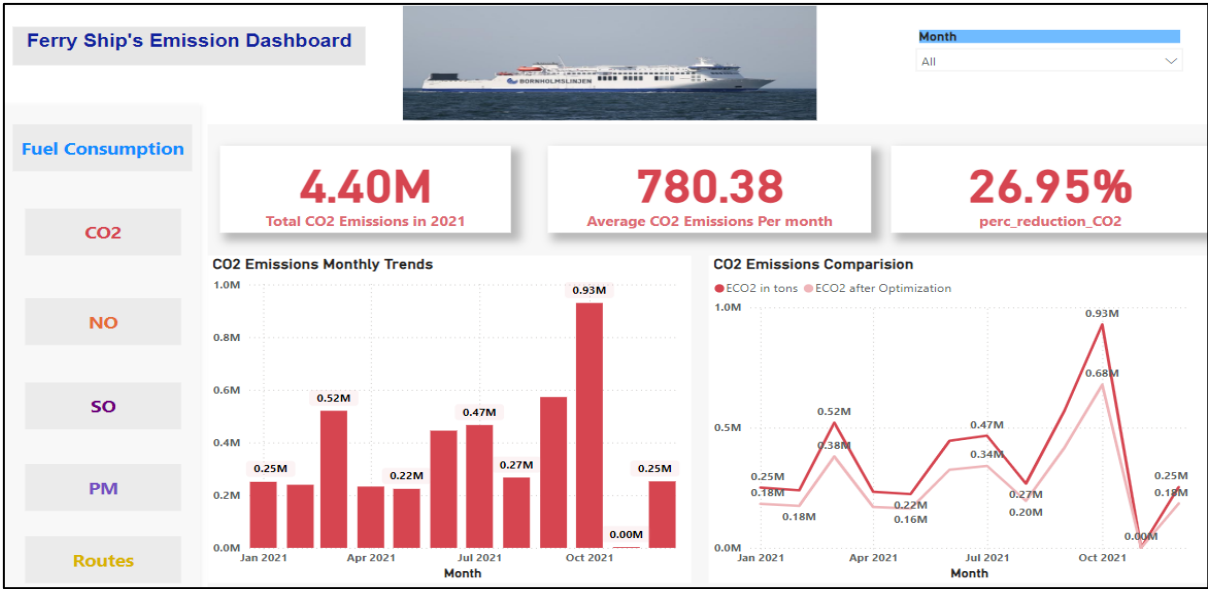


Figure 3. 6 Emission Dashboard (Source: Own illustration, 2024)

4 Results

The results section presents a thorough examination of the emissions produced by the ferry ship, including carbon dioxide (CO₂), nitrogen oxides (NO_x), Sulphur oxides (SO_x), particulate matter (PM), and fuel consumption. By employing the Holtrop-Mennen approach and utilizing Power BI for visualization, we obtained comprehensive insights into the environmental impact of the ferry ship's operations.

Fuel Consumption:



Figure 4. 1 Total fuel consumption in tonnes / kwh (Source: Own illustration, 2024)



Figure 4. 2 Average fuel consumption per month in tonnes/kwh (Source: Own illustration, 2024)

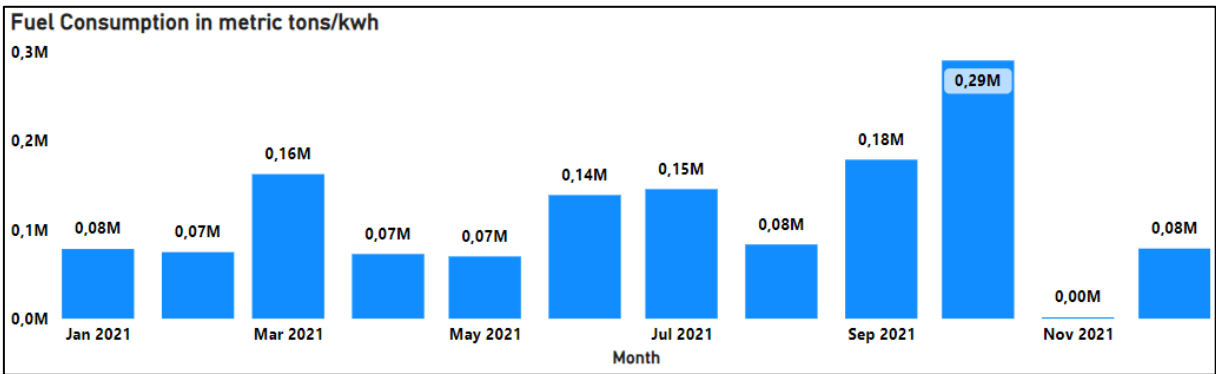


Figure 4. 3 Fuel consumption per each month in tonnes/kwh (Source: Own illustration, 2024)

Real-time and interactive fuel usage analysis is available from the fuel consumption page in Power BI regarding the ferry ship. A total fuel consumption parameter is, thus, a total summary of use, and an average fuel consumption parameter standardizes these data for comparison of efficiencies. Besides, the monthly fuel use in terms of bar chart is a good method to show the usage pattern, hence, stakeholders can easily notice the trend and change. The interactive capabilities enhance user-friendliness for informed decision-making in maximizing fuel efficiency and supporting environmental sustainability in maritime activities.

CO2 Emissions:



Figure 4. 4 Total CO2 Emissions in tonnes (Source: Own illustration, 2024)



Figure 4. 5 Average CO2 Emissions per month in tonnes (Source: Own illustration, 2024)

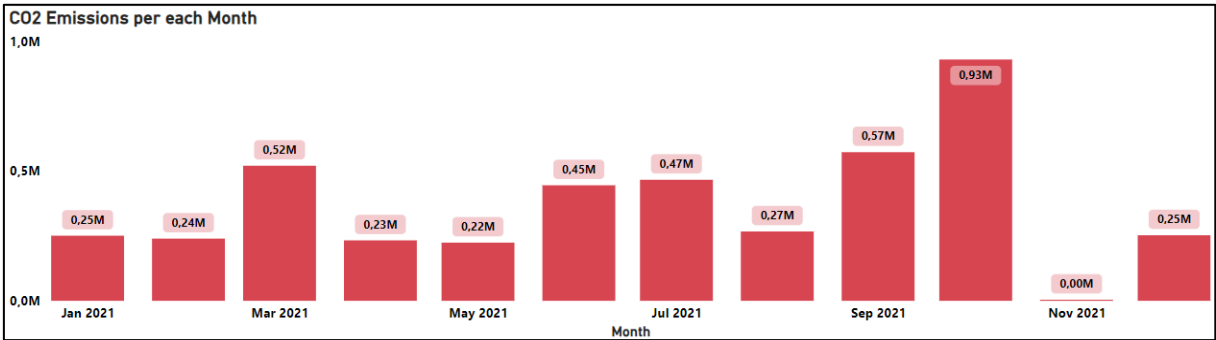


Figure 4. 6 CO2 Emissions per each month in 2021 (Source: Own illustration, 2024)

The CO2 emissions page in Power BI presents full details on the emissions in carbon dioxide of the ferry ship. In an interactive graphic manner, it shows shareholders a full picture of the volumes of CO2 emissions on the scale and over time. The total CO2 emissions metric provides a vivid picture of the ship's carbon footprint, while interactivity allows the user to study shorter timeframes individually or narrow the data down for detailed study. The main goal of this page is to improve decision-making in maritime operations to the point where it gives comprehensive information on the dynamics of CO2 emissions. This will enable the users to make informed decisions aimed at the environmental footprint and related issues, thus promoting sustainability.

NO Emissions:



Figure 4. 7 Total NO Emissions in tonnes (Source: Own illustration, 2024)

632.87

Figure 4. 8 Average NO Emissions per month in tonnes (Source: Own illustration, 2024)

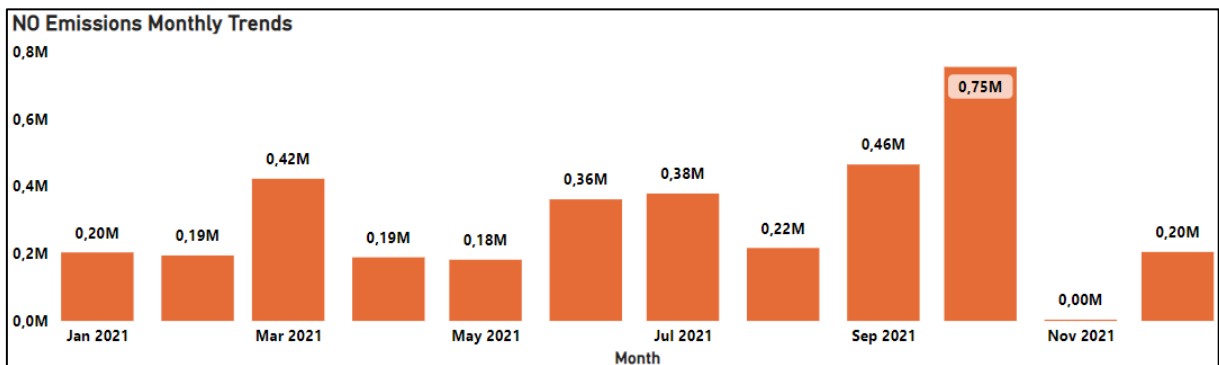


Figure 4. 9 NO Emissions per each month in tonnes (Source: Own illustration, 2024)

The NO_x emissions page of the Power BI provides a detailed analysis of nitrogen oxide emissions from the ferry ship. In using dynamic visualizations, stakeholders can easily see and understand patterns and trends in NO_x emissions over a period.

The total NO_x emissions measure provides an overview of the effect of the ship on air pollution. Interactive features make it possible to study changes and dependencies on operational parameters. This page provides information on the dynamics of NO_x emissions that might be useful in the implementation of reduction methods in connection with good air quality in maritime transport.

SO Emissions:

1.51K

Figure 4. 10 Total SO Emissions in 2021 in tonnes (Source: Own illustration, 2024)

0.27

Figure 4. 11 Average SO Emissions per month in tonnes (Source: Own illustration, 2024)

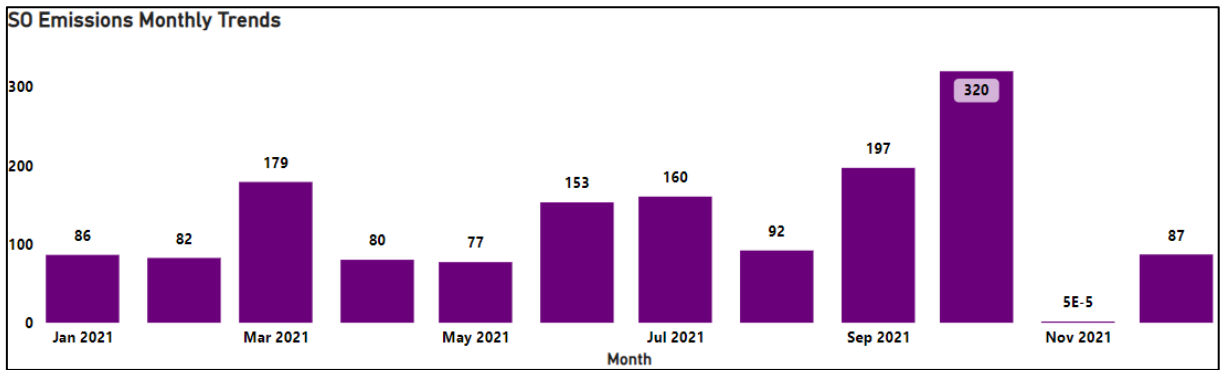


Figure 4. 12 SO Emissions per each month in 2021 (Source: Own illustration, 2024)

The Power BI page for SOx emissions provides information and details regarding the levels of Sulphur oxide emissions that are coming directly from the ferry ship. Stakeholders will use interactive visualizations to monitor and analyses trends in variations of SOx emissions over a range of time.

The total SOx emissions statistic quantifies the amount of Sulphur oxide emissions produced by the vessel, while interactive elements enable more investigation and examination. This page pushes for attempts to reduce Sulphur oxide emissions and thereby also help reduce negative impacts from the emissions on air quality and ecosystems, by underlining the fact that the SOx emissions have on environmental and human health.

PM Emissions:

247.16K

Figure 4. 13 Total PM Emissions in 2021 in tonnes (Source: Own illustration, 2024)

43.81

Figure 4. 14 Average Emissions per month in tonnes (Source: Own illustration, 2024)

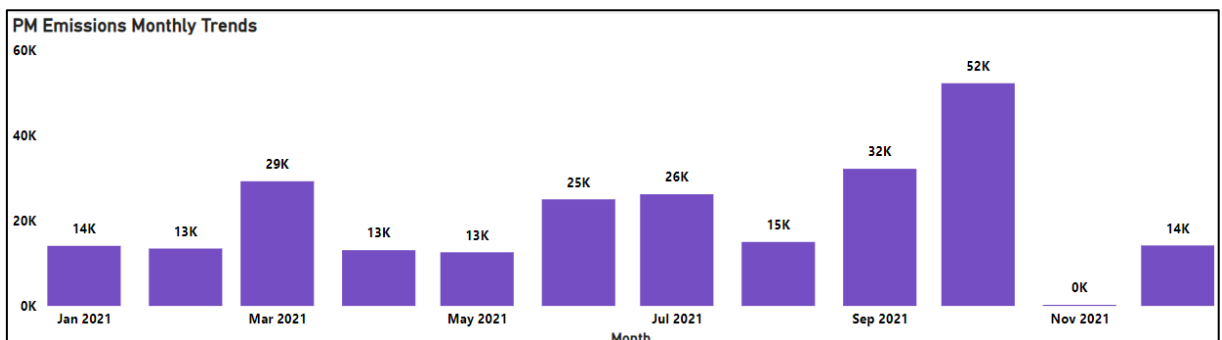


Figure 4. 15 PM Emissions in tons per each month in 2021 (Source: Own illustration, 2024)

The PM Emissions Power BI page facilitates the analysis of particulate matter emissions directly associated with the ferry vessel. Interactive graphs make it possible to identify recurring patterns and changing tendencies in particulate matter (PM) emissions over time. The overall PM emissions measure gives a comprehensive evaluation of the particulate matter pollutants emitted by the ship in question and allows for a more detailed analysis through the interactive features. On the PM Emissions page, information about the environmental and health impacts of PM emissions is provided to assist efforts that work towards reducing particulate matter emissions and maintaining good air quality in maritime transport.

Factors influencing Fuel Consumption and Emissions:

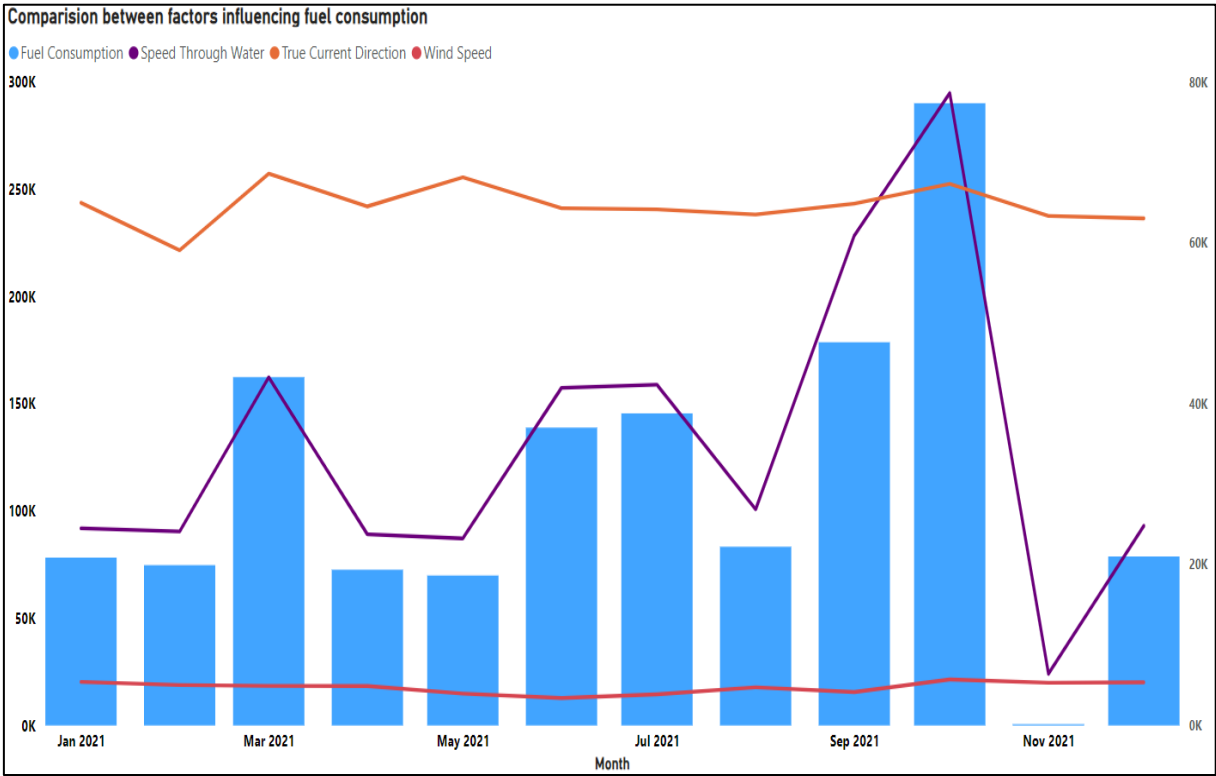


Figure 4. 16 factors influencing fuel consumption and emissions (Source: Own illustration, 2024)

The line chart on the fuel consumption tab of the Power BI dashboard illustrates the correlation between speed and fuel usage for the ferry ship. The data clearly demonstrates a direct association between speed and fuel consumption, highlighting speed as the key factor affecting fuel use. Changes in ship velocity directly correlate with fluctuations in fuel consumption, as higher speeds necessitate more fuel consumption for propulsion. Although the visualization includes wind speed and genuine wind wave direction, the observed data suggests that their influence on fuel use is minor or nonexistent. Although wind speed and real wind wave direction are present on the figure, there is no significant correlation

between these factors and variations in fuel usage during the observation period. The choice to exclusively concentrate on fuel use in the line chart is in accordance with its direct correlation to emissions, as fuel consumption is a key factor in determining emissions output. By solely examining fuel usage, stakeholders can obtain a comprehensive picture of the emissions characteristics of the ferry ship and pinpoint areas where emissions can be reduced, and fuel efficiency can be improved.

Fuel Consumption and Emissions Reduction:

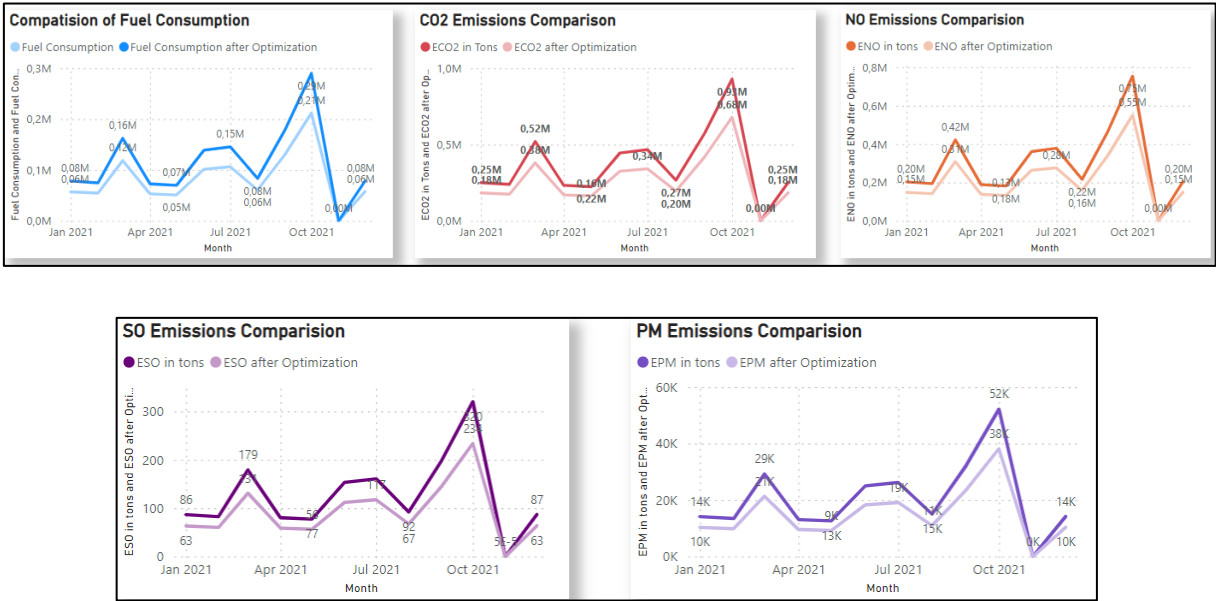


Figure 4. 17 Comparison of fuel consumption and emissions after optimization (Source: Own illustration, 2024)

The implementation of a 10% reduction in the speed of the ferry ship resulted in a substantial 29.65% drop in fuel consumption and emissions, including CO2, SO, NO, and PM. This was an operational change that strategically exploited a non-linear relationship between speed, fuel consumption, and emissions. In effect, it would achieve improved operational efficiencies by slowing down for a major economic benefit and environmental gain.

This decrease reflects the effectiveness of proactive measures in optimizing fuel efficiency and ensuring sustainability in maritime operations.

Journey Frequency and Emissions Mapping:

It contains a monthly display of the number of ferry trips, which really encapsulates an overall and complete insight into the operational trends throughout time.

It also provides a map visualization of latitude and longitude coordinates on ferry trips, giving stakeholders an understanding of spatial routes traveled.

This is in addition to the bar charts for users to delve into their individual journey emissions. Users can select a specific journey and look at its emissions profile, which allows one to make

specific analysis to support decisions aimed at achieving operational efficiency with the least environmental impact.

In the following page in Power BI, the month of January has been selected to highlight, including a total of 66 journeys. Users can select a particular journey to have the emission performance of the ship displayed.

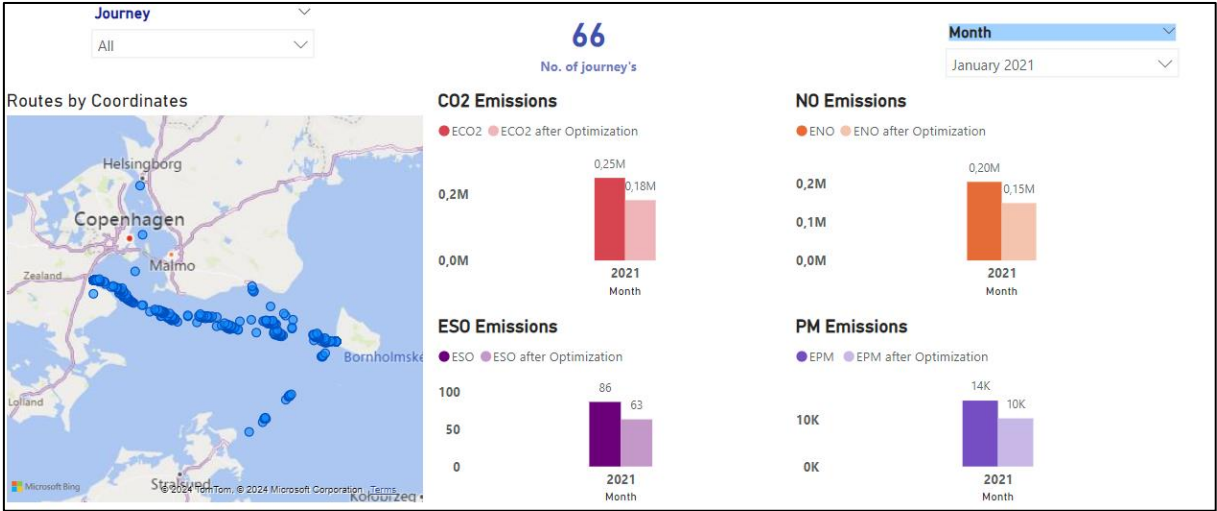


Figure 4. 18 Journey frequency and emissions mapping in January (Source: Own illustration, 2024)

5 Conclusion

The major aim of the present study was to develop a decision support system for accurate prediction of emissions from the ferry ship Hammershus. Using AIS data from 2021, a Python script was integrated into Power BI to compute emissions in the vessel following the Holtrop-Mennen method. This has, therefore, enabled us to gain an overall picture of the environmental impact associated with ferry operations, which in turn facilitates the decision process regarding sustainability.

A study into emissions dynamics has revealed a very complex relationship between operational factors and environmental results for the ferry ship Hammershus. Using the Python scripting in combination with Power BI analytics, stakeholders have been able to derive practical insights into determinants of emissions emanating from vessel speed and precise route details based on AIS data.

Besides, targeted reduction in the speed of the ferry ship Hammershus by 10% has successfully demonstrated that proactive measures can lead to tangible reduction in fuel consumption and emissions by 27%. The use of analytics and visualization tools has enabled the ferry operators to identify areas for optimization and implement measures that lower the environmental impact of the Hammershus.

Ease of visualization in showing the frequency of the journeys and mapping of the emissions provides stakeholders with in-depth knowledge on the ferry operations and emission characteristics unique to the Hammershus. The interactivity in exploring data on emissions specific to a vessel has supported informed decision-making to enhance operation efficiency and minimize environmental impact.

Finally, this study supports the implementation of decision support systems and data-driven methods to enhance environmental decision-making in ferry ship operations. More than that, it targets Hammershus ferry ship. The strategic use of AIS data, Python scripting, and Power BI analytics enables ferry operators to predict the emissions, identify hotspots for high emissions, and implement targeted strategies to reduce emissions from the ships. The Hammershus is leading the way toward greener ferry operations within the marine industry: greener because of this cooperation and technical innovation, now and in the future. This promises a cleaner and more sustainable future.

5.1 Future Recommendations

In the future, multiple options exist to enhance the environmental sustainability of ferry operations, especially for the ferry vessel Hammershus. Real-time monitoring systems of emissions will be of value since they provide valuable information, thereby allowing for direct alterations of operational means in a way that achieves maximum fuel utilization and minimum impact on the environment. In addition, increased sophistication of the prediction models can avail forward-looking recommendations of means towards emission reduction. This is achievable via the use of historical data and machine learning algorithms to enhance the level of accuracy. Further, it is essential to explore opportunities for the use of renewable energy and encourage collaboration in the maritime industry to enhance the achievement of the sustainability goals. Continuous research, meaningful participation of the stakeholders, and educational programs will ensure a high level of commitment to environmental management.

Eventually, with the implementation of these future suggestions and adopting a sustainable mindset, the ferry Hammershus will be at the frontline in ensuring that it plays a role in saving the world from an unpredictable future. This will make the Maritime Industry have a sustainable and cleaner environment and thus serving as a perfect example for the coming generations.

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Erklärung

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